



Medical image identification methods: A review

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ABSTRACT

The identification of medical images is an essential task in computer-aided diagnosis, medical image retrieval and mining. Medical image data mainly include electronic health record data and gene information data, etc. Although intelligent imaging provided a good scheme for medical image analysis over traditional methods that rely on the handcrafted features, it remains challenging due to the diversity of imaging modalities and clinical pathologies. Many medical image identification methods provide a good scheme for medical image analysis. The concepts pertinent of methods, such as the machine learning, deep learning, convolutional neural networks, transfer learning, and other image processing technologies for medical image are analyzed and summarized in this paper. We reviewed these recent studies to provide a comprehensive overview of applying these methods in various medical image analysis tasks, such as object detection, image classification, image registration, segmentation, and other tasks. Especially, we emphasized the latest progress and contributions of different methods in medical image analysis, which are summarized base on different application scenarios, including classification, segmentation, detection, and image registration. In addition, the applications of different methods are summarized in different application area, such as pulmonary, brain, digital pathology, brain, skin, lung, renal, breast, neuromyelitis, vertebrae, and musculoskeletal, etc. Critical discussion of open challenges and directions for future research are finally summarized. Especially, excellent algorithms in computer vision, natural language processing, and unmanned driving will be applied to medical image recognition in the future.

1. Introduction

With the development of medical imaging analysis technology, medical images are becoming more and more important in clinical diagnosis, in which medical image analysis plays an indispensable role in medical diagnosis for clinical therapy [1–4]. Image processing technology techniques can perform qualitative or quantitative analysis of specific areas, which can greatly improve the efficiency and accuracy of clinical diagnosis. Not only that, medical images can also play a key role in medical teaching, surgical planning and so on. The tasks of medical images are mainly divided into three categories: classification and recognition, localization and detection, and segmentation. These three types of tasks have been widely used in clinical auxiliary screening, diagnosis, grading, treatment decision-making and guidance, and efficacy evaluation of benign and malignant tumors, brain function and mental disorders, cardiovascular and cerebrovascular diseases and other

major diseases. Deep learning models can discover complex structures in high-dimensional data, which has achieved amazing results in the field of medical images [5]. Intelligent imaging and machine learning are mainly used to solve the needs of different image diagnosis, such as image segmentation, feature extraction, quantitative analysis and comparative analysis of X-ray, computed tomography (CT), magnetic resonance imaging (MRI), and data identification and annotation. At the same time, the images analysis with machine learning can reduce the incidence of false negative diagnosis and improve the efficiency of film reading. Intelligent imaging and machine learning can also carry out automatic delineation and other image processing mainly for tumor radiotherapy, and continuously identify the change of the lesion position during the patient's radiotherapy, so as to realize adaptive radiotherapy and reduce the radiation to healthy tissues.

With the rapid development of artificial intelligence, image processing methods have achieved good results in the field of image

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segmentation. Compared with traditional computer vision methods, machine learning has certain advantages in segmentation accuracy and speed. Therefore, the use of machine learning to segment medical images can effectively help doctors confirm the size of diseased tumors, quantitatively evaluate the effect before and after treatment, greatly reducing the workload of doctors [6,7]. Image identification was one of the first areas for medical imaging analysis. Both medical image classification and segmentation belong to the category of medical image recognition. Medical image classification aims to distinguish medical images according to a certain criterion including imaging modalities or clinical pathologies. A reliable medical image classification system is

able to assist doctors in the fast and accurate interpretation of medical images [8,9]. Medical image segmentation mainly deals with the segmentation problems of various images involved in the medical field, such as common MRI scanning images. The main task of medical image segmentation is to segment regions of interest from these medical images including specific organ parts, objects of interest, organs segmentation and other substructures in medical. Identifying the set of voxels is typically defined as the task of segmentation in medical imaging analysis. The voxels make up either the contour or the interior of the interested object. Segmentation is the most common subject of papers applying deep learning [10–13]. Fig. 1 showed key contributions for

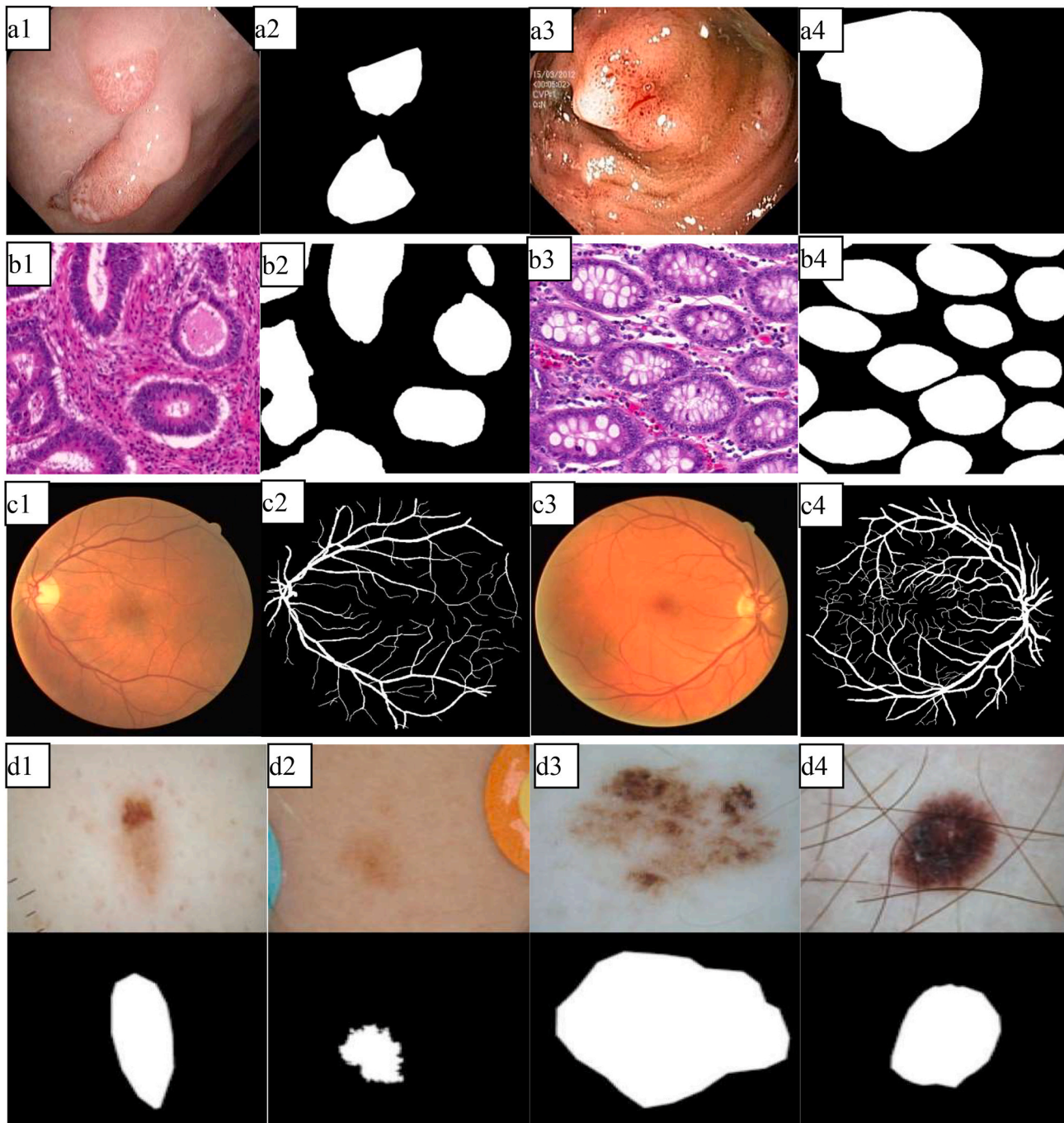


Fig. 1. Medical image classification and segmentation on large data sets and on public challenge data sets. (A) Polyp segmentation dataset Kvasir-SEG (a1-4) (<https://datasets.simula.no/kvasir-seg/>); (B) eye vessel segmentation dataset DRIVE (b1-4) (<https://drive.grand-challenge.org/>); (C) breast cancer cell segmentation dataset GLAS (c1-4) (<https://bioimage.ucsb.edu/research/bio-segmentation>); (D) skin disease segmentation dataset ISIC 2018 (d1-4) (<https://challenge.isic-archive.com/landing/2018/>).

medical image classification and segmentation on large data sets and on public challenge data sets including Polyp segmentation dataset Kvasir-SEG; eye vessel segmentation dataset DRIVE; breast cancer cell segmentation dataset GLAS; skin disease segmentation dataset ISIC 2018.

Medical image classification has been thoroughly studied during the past decades. Most of the literatures are based on handcrafted features. Despite the success of these methods, handcrafted features with optimal specific classification task are usually difficult to design. Some new image classification techniques including machine learning (ML), deep learning (DL), convolutional neural networks (CNN), transfer learning (TL) is summarized in this paper. Machine learning has been praised as one of the top ten technological breakthroughs since 2013 due to its considerable application prospects in data analysis. Before deep learning, letting computers complete the task of image recognition was still dominated by classical machine learning algorithms. Machine learning [14–16] derived from artificial intelligence [7,17–22] and statistics explores learning, which analyzes data outside the training dataset with specific models. In particular, machine learning plays an important role in image processing. Subsequently, a large number of researchers have used machine learning to recognize and segment medical images.

And soon after, deep learning techniques, especially deep convolutional neural networks (DCNN) [23,24], have led to significant breakthroughs in medical image classification and segmentation [25–27]. The deep learning method simulates the human neural network. By combining multiple nonlinear processing layers, the original data is abstracted layer by layer, and different levels of abstract features are obtained from the data and used for target detection, classification or segmentation. The advantage of deep learning is to replace the manual acquisition feature with unsupervised or semi-supervised feature learning and hierarchical feature extraction efficient algorithms. Specifically, the task of medical image classification and segmentation was carried out around the MRI plain scan images of the heart, CT images of the brain and chest X-ray images of the lung, which the diagnostic and application value of the deep learning model was explored [28–30]. Deep learning methods learn the fusion model with good generalization ability from a large amount of data, which can make the fusion process more robust and overcomes the shortcomings of manual feature selection, such as expensive, time-consuming and prone to human errors. Su et al. [31] proposed a novel and practical method to detect gastrointestinal disease from wireless capsule endoscopy (WCE) images by DCNN. The proposed method utilized three backbone networks modified with transfer learning as the feature extractors. Gastrointestinal diseases were detected by an integrated classifier that is trained using ensemble learning. Deep learning also can significantly improve the efficiency of identifying skin disorders and alleviate the scarcity of medical resources. Zhou et al. [32] proposed a deep learning schema to study the influence of color-based background selection on a deep model's capacity. The novel model combined data preprocessing, data augmentation, and residual networks to learn foreground lesion subject attributes in a skin disease classification problem.

Convolutional neural network is one of the most prominent forms for the classical multi-layer neural network in medical image classification and segmentation [33–36]. CNN has become the workhorse DL algorithm for medical image classification and segmentation [37–40]. In recent data challenges for medical image analysis, all of the top-ranked teams utilized CNN [41]. The different of CNN and traditional fully-connected neural networks is that the CNN can maintain the correlation information of the surrounding data. However, CNN comes with many challenges in practical problems. CNN really made a breakthrough in the field of image recognition until the proposal of AlexNet [42], and it greatly surpassed the traditional machine learning for the international image recognition competition. Nowadays, CNN are shining in the field of not only image recognition but also computer vision. For example, a multiscale view (MV) module was proposed by Wang et al.

[43] to detect multi-scale brain network disorders. 3D multiscale view convolutional neural network with attention (3D MVA-CNN) is proposed to classify MRI images for mental disease, which promoted the assisted diagnosis of mental diseases. Shao et al. [44] explored whether the Nomogram could accurately predict placenta invasion combining the CNN and radiomic features of T2-weighted MR images.

Owing to the requirement of large amounts of data with detailed annotations in supervised learning, transfer learning has received a lot of attention. Many research endeavors have tried to overcome data scarcity problem with TL or domain adaptation techniques. These aim to achieve high performance on target tasks by leveraging knowledge learned from source tasks. Pan et al. [45] contributed a pioneering review paper about TL, in which TL techniques is classified from a labeling aspect. Weiss et al. Zhuang et al. [46] reviewed more than forty representative TL approaches from the perspectives of data and models. The aim is to identify research directions, gaps in knowledge, applications, and widely used strategies among the TL approaches applied in MR brain imaging. Valverde et al. [47] proposed the first systematic review of transfer learning in MR brain imaging applications. The review highlighted brain-specific methods among the TL applying in MR brain imaging. Agarwal et al. [48] introduced a systematic review of the current state of early Alzheimer's disease (AD) detection by using TL and neuroimaging biomarkers. Others surveyed articles focusing on TL with a specific case study such as cervical cytopathology and microorganism counting. Rahaman et al. [49] introduced a comprehensive study of the state of the art approaches based on TL for the analysis of cervical cytology images. The publicly available cervical cytopathology datasets and evaluation metrics for segmentation and classification tasks are discussed. Zhang et al. [50] considered the development of microorganism counting methods using digital image analysis, in which image processing methods for microorganism counting are summarized and analyzed.

Although some review on application of medical image analysis covers a substantial amount of work, we feel that important areas were not represented. For instance, no work on retinal image analysis was covered. The motivation for our review was to offer a comprehensive overview of (almost) all fields in medical imaging, both from a methodology driven and an application perspective. This also includes overview tables of all publications which readers can use to quickly assess the field. Last, we leveraged our own experience with the application of different image identification methods to provide up-to-date and comprehensive knowledge covering the state-of-the-art, open challenges and overview of research directions and technologies that will become important in the future.

This survey includes over 186 papers, most of them recent, on a wide variety of applications of several well-known techniques for medical image classification and segmentation. To identify relevant contributions, web of science was queried for papers containing ("medical image classification" or "medical image segmentation") in title or abstract. ArXiv was searched for papers mentioning one of a set of terms related to medical imaging. References in all selected papers are checked. The publication(s) deemed most important were included when overlapping work had been reported in multiple publications. We expect the search terms used to cover most, if not all, of the work incorporating convolutional neural network, deep learning, machine learning, transfer learning, and image processing technology. The last update to the included papers was on May 1, 2023.

Summarizing, with main contribution of this survey we aim to:

- Multiple representative medical image identification approaches are introduced and summarized, which can give readers a comprehensive overview about medical image identification approaches;
- Show that different techniques including convolutional neural network, deep learning, machine learning, transfer learning, and image processing technology, have permeated the entire field of medical image identification;

- Actionable recommendations on how to leverage medical image identification methods for medical image classification and segmentation are provided;
- Identify the challenges for successful application of these techniques to medical imaging tasks;

The remainder of this survey is structured as followed. In Section 2, we introduced medical image identification research studies and publication analysis. Section 3 described application of different techniques including convolutional neural network, deep learning, machine learning, transfer learning, and image processing technology in medical image recognition. Meanwhile, open challenges in different application areas: ophthalmic, neuron, brain, pulmonary, chest, cardiac, breast, head, abdominal, bone, musculoskeletal, and remaining miscellaneous applications, are discussed. Section 4 presented a summary and an outlook for future research.

2. Research studies and publication analysis on medical image identification

Image classification and segmentation was one of the first areas in which medical image identification methods made a major contribution for medical image analysis [51–54]. 74826 articles related to medical image analysis studies have been published in journals/dissertations/conferences up to May 2023 since 2014. The number of papers grew rapidly in 2019 and 2022. Temporal histogram of the collected published articles is illustrated in Figs. 2–3.

The collected articles are from a wide variety of journals covering the research fields of radiology nuclear medicine medical imaging, mathematical computational biology, computer sciences, engineering, communication, mathematics, neurosciences neurology, geriatrics gerontology, imaging science photographic technology, science technology other topics, biochemistry molecular biology, telecommunications, oncology, research experimental medicine, cardiovascular system cardiology, health care sciences services, automation control systems.

Among these 74826 papers, 51046 papers were published in the radiology nuclear medicine medical imaging field, accounting for 68.220 %; 45843 papers were published in the mathematical computational biology field, accounting for 60.841 %; 39135 papers were published in the engineering field, accounting for 52.301 %; 36924 papers were published in the communication field, accounting for 49.346 %; 27.762 papers were published in the mathematics field, accounting for 37.102 %; 18085 papers were published in the neurosciences neurology field, accounting for 24.169 %; 13520 papers were published in the science technology of other topics field, accounting for

18.069 %; 11115 papers were published in the oncology field, accounting for 14.854 %; 7717 papers were published in the cardiovascular system cardiology field, accounting for 10.313 %; 7590 papers were published in the geriatrics gerontology field, accounting for 10.144 %, as shown in Figs. 4–5.

3. Application of different techniques on medical image recognition

In this section, we selected 186 articles related to medical image classification and segmentation from 74826 articles, which have high citation and representative. Most of them were recently published and wide applied to different learning techniques in medical image recognition. To identify relevant contributions web of science was queried for papers containing (“Medical image classification” or “medical image segmentation”) in title or abstract. Papers that not report results on medical image data or only used standard feed-forward neural networks with handcrafted features have been excluded. Only the publication(s) deemed most important were included. In addition to these, the concepts pertinent of methods, such as the machine learning, deep learning, convolutional neural networks, concepts pertinent of transfer learning, and image processing technology for medical image are reviewed to analyze and summarize in this section. We provided the background knowledge of different techniques, including the basic structure of these techniques and application literature for image representation.

3.1. Medical image identification used in convolutional neural network

3.1.1. Convolutional neural network

Convolutional neural network is a further in-depth multi-layer neural network which is known nowadays was proposed by LeCun et al. [55]. The idea of deep learning is introduced into the neural network. Features of different levels of the image are extracted through the convolution operation. The parameters of the convolution kernel is automatically adjust, which produce the most suitable classification features without supervision. CNN architecture can not only enhance the fitting effect between layers but also reduce redundant computations in the network, which can bring more superior performance. From data besides being trained as a classifier, CNN has the property of extracting features automatically.

Architecture representation of typical CNN usually consists of various combinations of convolutional layer, pooling layer, and fully connected layer. The convolutional layers are composed of multiple two-dimensional matrices, which is update using back propagation. Filters slide horizontally and vertically to scan the input images in this

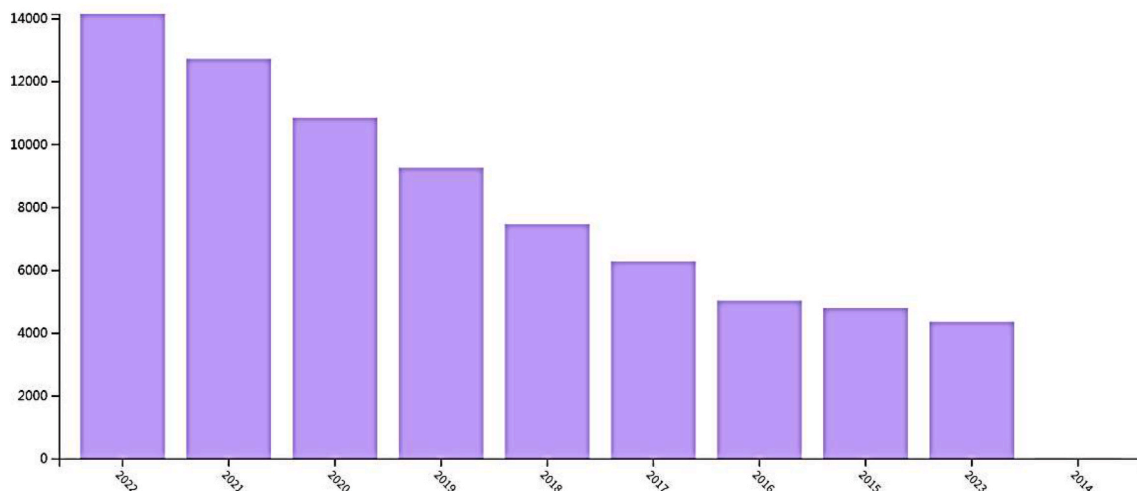


Fig. 2. Temporal histogram of the collected articles which been published in journals/dissertations/conferences up to May 2023 since 2014.

Select All ☐	Field: Publication Years	Record Count	% of 74,826
☐	2022	14,144	18.903%
☐	2021	12,716	16.994%
☐	2020	10,842	14.490%
☐	2019	9,250	12.362%
☐	2018	7,457	9.966%
☐	2017	6,271	8.381%
☐	2016	5,024	6.714%
☐	2015	4,789	6.400%
☐	2023	4,352	5.816%
☐	2014	4	0.005%

Fig. 3. Ranking with a wide variety of the collected published articles including record count and percentage.



Fig. 4. A wide variety of journals covering the research fields of radiology nuclear medicine medical imaging in 74826 papers.

layer. The input image is multiplied with the weight matrices filters in convolution operation. After convolution operation, the pooling layer is used to reduce the feature value of the input matrixes. The architecture of CNN can be shown in Fig. 6.

Convolution layer aims to slide a predefined fixed-size window on the feature map. Each feature tiles at various positions are extracted adjacent step by step, and obtained vector space is reorganized to acquire a new tensor. The input and output of convolution operation is Eqs. (1) and (2):

$$x_j^l = f^l(u_j^l) \quad (1)$$

$$u_j^l = \sum_{i \in M_j} x_i^{l-1} * k_{ij}^l + b_j^l \quad (2)$$

where x_j^l and u_j^l express the net activation and output for j th channel in the convolution layer l , respectively. $f^l(\cdot)$ is the activation function of

convolution layer l . M_j expresses the feature map subset sampled. b_j^l and k_{ij}^l represent the bias of feature map and the convolution kernel matrix of convolution layer l , respectively.

Pooling layer named as the down-sampling layer includes carpooling, mean-pooling and stochastic pooling, which takes the maximum value, mean value, and random feature points of feature points in the neighborhood, respectively. The input and output of pooling operation is shown in Eqs. (3) and (4):

$$x_j^l = f^l(u_j^l) \quad (3)$$

$$u_j^l = a_j^l \cdot f_{pooling}^l(x_j^{l-1}) + b_j^l \quad (4)$$

where x_j^l and u_j^l express the net activation and output for j th channel in the pooling layer l , respectively. $f^l(\cdot)$ is the activation function of pooling

Select All ☐	Field: Research Areas	Record Count	% of 74,826
☐	Radiology Nuclear Medicine Medical Imaging	51,046	68.220%
☐	Mathematical Computational Biology	45,843	61.266%
☐	Computer Science	45,525	60.841%
☐	Engineering	39,135	52.301%
☐	Communication	36,924	49.346%
☐	Mathematics	27,762	37.102%
☐	Neurosciences Neurology	18,085	24.169%
☐	Science Technology Other Topics	13,520	18.069%
☐	Oncology	11,115	14.854%
☐	Cardiovascular System Cardiology	7,717	10.313%
☐	Geriatrics Gerontology	7,590	10.144%

Fig. 5. Ranking with a wide variety of journals covering the different research application areas for medical image in 74826 papers.

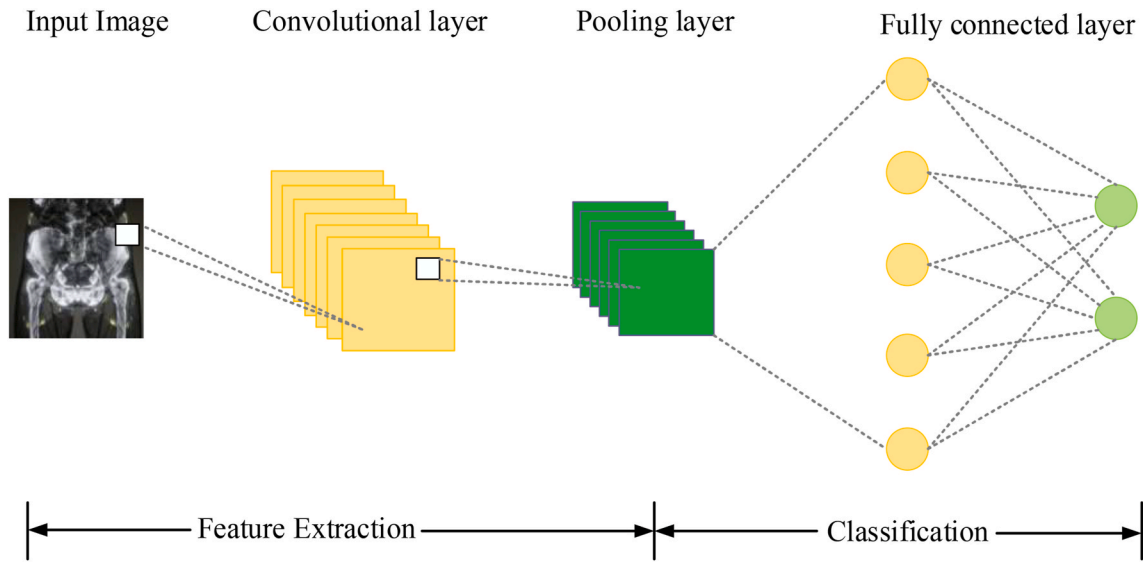


Fig. 6. Representation of convolutional neural network architecture.

layer l . $f_{pooling}^l$ is the pooling function in the pooling layer l . a_j^l stands for denotes the weight coefficient of pooling layer l .

Fully connected layer is the last part of a convolutional neural network. The two-dimensional (2D) feature information output by the previous layer is converted into the one-dimensional (1D) classification information. The input and output of fully connected operation is shown in Eqs. (5) and (6):

$$x_j^l = f^l(u_j^l) \quad (5)$$

$$u_j^l = \max\left(0, \sum_i y_i^l \cdot w_{ij}^l + b_j^l\right) \quad (6)$$

where x_j^l and u_j^l express the net activation and output for j th channel in the fully connected layer l , respectively. $f^l(\cdot)$ is the activation function of

fully connected layer l . $\max(\cdot)$ is largest number. b_j and w_{ij}^l express bias and the weight coefficient of fully connected layer l , respectively.

3.1.2. Convolutional neural network applied on medical images

Convolutional neural network is one of the most widely techniques in medical images, especially the field of medical image analysis. We summarize the recent literature using convolutional neural network technology for seven parts of the human body. Among them, 19 papers were published for the chest (cardiomegaly, lung, heart, liver, gastric) parts of the human body, accounting for 40 %. 12 papers were published for the brain parts of the human body, accounting for 25 %. 5 papers were published for the breast and cervical parts of the human body, accounting for 11 %. 4 papers were published for the head (eye, nose) parts of the human body, accounting for 8 %. 4 papers were published for the Skin parts of the human body, accounting for 8 %. 2 papers were published for the vertebrae parts of the human body, accounting for 4 %.

2 papers were published for the abdomen parts of the human body, accounting for 4 %, as shown in Fig. 7. Review of head, skin, chest, brain, vertebrae, abdomen and breast using convolutional neural network are shown in Tables 1–6, respectively.

Wan et al. [56] designed a magnetic resonance imaging classification algorithm by using convolutional neural network, which accurately classified cerebral small vessel disease and improved the classification performance. Early diagnosis of Alzheimer's disease is critical. Ashtari-Majlan et al. [57] developed a multi-stream deep convolutional neural network to classify stable mild cognitive impairment. These landmarks with multivariate statistical test are used to extract patches. These landmarks are fed into multi-stream convolutional neural network for classifying magnetic resonance imaging.

The authors in Ref. [58] introduced a novel model which combined depth-wise separable convolutions with the Bayesian algorithm for predictions and classification of brain tumors. The novel model provided accurate prediction results for the magnetic resonance imaging images. A transfer learning densely connected convolutional dictionary learning framework (TLD-CDL) is proposed to remove noise and artifacts from low-dose computed tomography (LDCT) in Ref. [59]. The denoised results (Case 5 slice) of the ablation experiments on Mayo dataset, is shown in Fig. 8. The integration of attention modules is presented in Ref. [60] to develop a robust multi-path network, and then its multiple feature-extraction blocks is leveraged for medical image (dementia grading tasks, brain disorder classification, and brain tumor classification) non-hierarchical mining. Three models based on convolutional neural network architecture is introduced to segment the cerebellar fissures for brain magnetic resonance imaging in Ref. [61]. The performance results showed that the obtained model outstanding performance in terms of efficiency and precision. Shi et al. [62] developed a novel residual neural network for stroke classification. The authors quantitatively evaluated the proposed method in three different layers.

A new CNN based on the convolutional U-net architecture [63] is introduced to pre-process the magnetic resonance images. The numerical results indicated that the propose method obtained excellent classification accuracy for magnetic resonance medical images. In Ref. [64], Alijamaat et al. applied the Haar wavelet transform to make use of the spectral information, in which the two-dimensional discrete Haar wavelet transform and the CNN network is combined. 676 multiple sclerosis brain slices and 681 healthy control brain slices are collected

Table 1
Review of brain pathologies detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Wan et al. [56]	2023	Brain	Cerebral small vessel disease	Convolutional neural network
Ashtari-Majlan et al. [57]	2022	Brain	Alzheimer's disease diagnosis	Multi-stream deep CNN
Ekong et al. [58]	2022	Brain	Brain tumor MRI classification	Bayesian depth-Wise CNN
Yan et al. [59]	2023	Brain	Low-dose CT Image denoising	Convolutional dictionary learning Deep CNN
Apostolopoulos et al. [60]	2023	Brain	Brain tumor classification	Convolutional neural network
Cabeza-Ru et al. [61]	2022	Brain	Cerebellum segmenting	Residual CNN
Shi et al. [62]	2022	Brain	Stroke classification	
Hashemzahi et al. [63]	2021	Brain	Brain tumors classification	CNN based on U-net architecture
Alijamaat et al. [64]	2021	Brain	Multiple sclerosis identification	Wavelet CNN
Zhang et al. [65]	2018	Brain	Multiple sclerosis identification	CNN with dropout and ReLU
Wang et al. [66]	2021	Brain	Multiple sclerosis identification	CNN with multiple-way
Soltani et al. [67]	2020	Brain	Multiple sclerosis identification	Convolutional neural network

and used to develop an improved convolutional neural network system which combined the parametric rectified linear unit (PReLU) and dropout techniques [65]. A 6-layer stochastic pooling convolutional neural network is introduced to detect multiple sclerosis in brain magnetic resonance imaging [66]. Soltani et al. [67] used convolutional neural network to detect multiple sclerosis (MS) disease lesions from MRI. The experiments showed that convolutional neural network has a strong potential for MS disease diagnosis, which provided good results without the need for lesions segmentation. Retinal blood vessels are importance significant for ophthalmologic disease diagnostic. Sule et al.

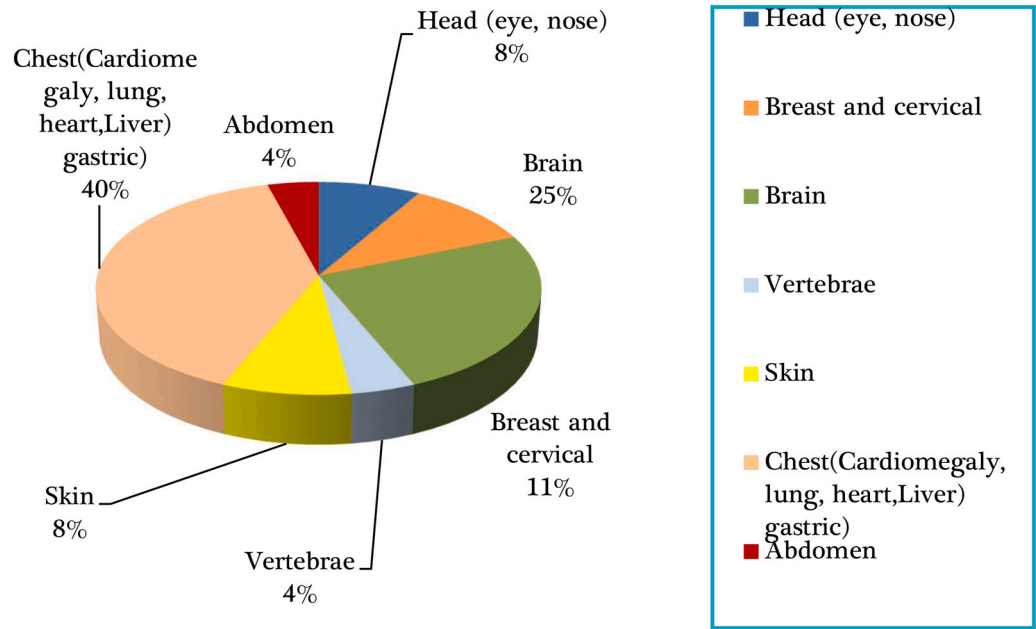


Fig. 7. Classification and proportion of medical image identification used in convolutional neural network.

Table 2
Review of head pathologies detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Sule et al. [68]	2022	Eye	Retinal fundus images segmentation	Improved CNN
Mohite et al. [69]	2022	Eye	Retinal fundus images segmentation	Multilayered CNN
Seeböck et al. [70]	2018	Eye	Retinal imaging identification	Convolutional neural networks
Ji et al. [71]	2022	Nasopharynx	Nasopharyngeal carcinoma diagnosis	Convolutional neural networks

Table 3
Review of abdomen pathologies detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Li et al. [72]	2023	Abdomen	Pancreatic cysts segmentation	Convolutional neural network
Huang et al. [73]	2022	Abdomen	Hepatobiliary tumors diagnose	Convolutional neural network

Table 4
Review of breast pathologies detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Das et al. [74]	2023	Breast	Breast cancer detection	Shallow CNN
Chen et al. [75]	2022	Breast	Breast tumor screening	An optimal multilayer architecture
Sani et al. [76]	2023	Breast	Breast cancer classification	Deep CNN
Rahman et al. [77]	2023	Breast	Breast cancer detection	Computational framework with CNN
Dey et al. [78]	2023	Breast	Breast cancer detection	CNN architectures

[68] devised an improved CNN for optimal segmentation of retinal fundus images blood vessels. Some robust contrast enhancement tools are explored to effectively improve the quality of the fundus images. In order to improve retrieval performance, Mohite et al. [69] proposed a deep convolutional neural network model, a framework based on multilayered convolutional neural network, for image retrieval and classification. Medical images are represented as deep features for retrieval purpose in the framework. The retrieval accuracy of the model is enhanced by using a data augmentation technique. An unsupervised identification of anomalies is presented by Seeböck et al. [70] to segment anomalies for markers in retinal optical coherence tomography (OCT) imaging. CNN is also utilized for nasopharyngeal carcinoma (NPC) early diagnosis by Ji et al. [71].

A convolutional neural network architecture, pyramid attention and pooling on convolutional neural network (PAPNet), is proposed by Li et al. [72] for segmentation of pancreatic cysts in abdominal. The segmentation performance is effectively improved utilizing a spatial pyramid pooling module. Huang et al. [73] devised a diagnostic method of microscopic images of liver and gallbladder tumors based on CNN. The numerical results indicated that the CNN remarkably improved the success rate of the treatment.

Breast cancer is one of the most common cancers which affect women in all the cancers afflicted women. Das et al. [74] developed computer-aided diagnosis to help for diagnoses of breast cancer. Transfer learning is used to feed the same collection of images into pre-trained convo. It was demonstrated that the new approach outperformed all other state-of art techniques. CNN are usually used to classify for automatic breast tumor screening in mammography

Table 5
Review of chest pathologies detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Pandit et al. [79]	2022	Lung	Lung cancer classification	Convolutional neural network
Xu et al. [80]	2022	Lung	Pulmonary tuberculosis detection	CNN with coordinate
Lin et al. [81]	2022	Heart	Cardiomegaly level screening	Multilayer CNN classifier
Lin et al. [82]	2022	Lung	Lung cancer classification	Fusion-based fuzzy CNN
Castiglione et al. [83]	2021	Lung	COVID-19 diagnosis	Optimized CNN
Laddha et al. [84]	2022	Lung	COVID-19 diagnosis	Deep adversarial CNN
Alshudukhi et al. [85]	2022	Lung	Tuberculosis classification	Automatic registration algorithm
Palomo et al. [86]	2022	Lung	Pneumonia detection	Convolutional neural network
Mijwil et al. [87]	2021	Lung	COVID-19 diagnosis	Convolutional neural network
Ahmad et al. [88]	2022	Liver	Liver segmentation	Lightweight CNN
Hassanzadeh et al. [89]	2021	Liver	Liver segmentation	Deep CNN
Chen et al. [90]	2021	Heart	Myocardial perfusion classification	Convolutional neural network
Lossau et al. [91]	2019	Heart	Coronary motion artifact recognition	Convolutional neural network
Liang et al. [92]	2018	Gastric	Gastric cancer segmentation	Convolutional neural network
Wang et al. [93]	2021	Gastric	Gastric cancer segmentation	Deeplab v3 + neural network
Nagao et al. [94]	2020	Gastric	Gastric cancer recognition	Transfer learning and CNN
Qu et al. [95]	2018	Gastric	Gastric pathology classification	Stepwise fine-tuning-based CNN
Liu et al. [96]	2020	Gastric	Gastric disease classification	Fine-tuning pre-trained CNN
Lui et al. [97]	2020	Gastric	Gastric lesions prediction	Convolutional neural networks

Table 6
Review of skin and vertebrae detection using convolutional neural network.

Reference	Year	Pathology	Modality	Proposed approach
Salih et al. [98]	2023	Skin	Skin lesion diagnosis	CNN based on genetic algorithm
Villa-Pulgarin et al. [99]	2022	Skin	Skin cancer diagnosis	CNN based on transfer learning
Hosny et al. [100]	2022	Skin	Skin classification	Residual deep CNN
Sharma et al. [101]	2022	Skin	Skin cancer diagnosis	Handcrafted features based deep CNN
Cai et al. [102]	2016	Vertebrae	Vertebrae recognition	CNN based on genetic algorithm
Hetherington et al. [34]	2017	Vertebrae	Spine identification	Deep CNN

examination. An optimal multilayer architecture including three convolutional layers, two pooling layers, a flattening layer, and a classification layer, is developed by Chen et al. [75] for automatic breast tumor screening. The research results showed promising performance for automatic breast tumor screening.

The authors introduced a novel deep neural network architecture based on a group CNN (G-CNN) with discrete cosine transform (DCT) and a special Euclidean (SE2) motion group. The novel technique is used for improving breast cancer classification, which achieved a high

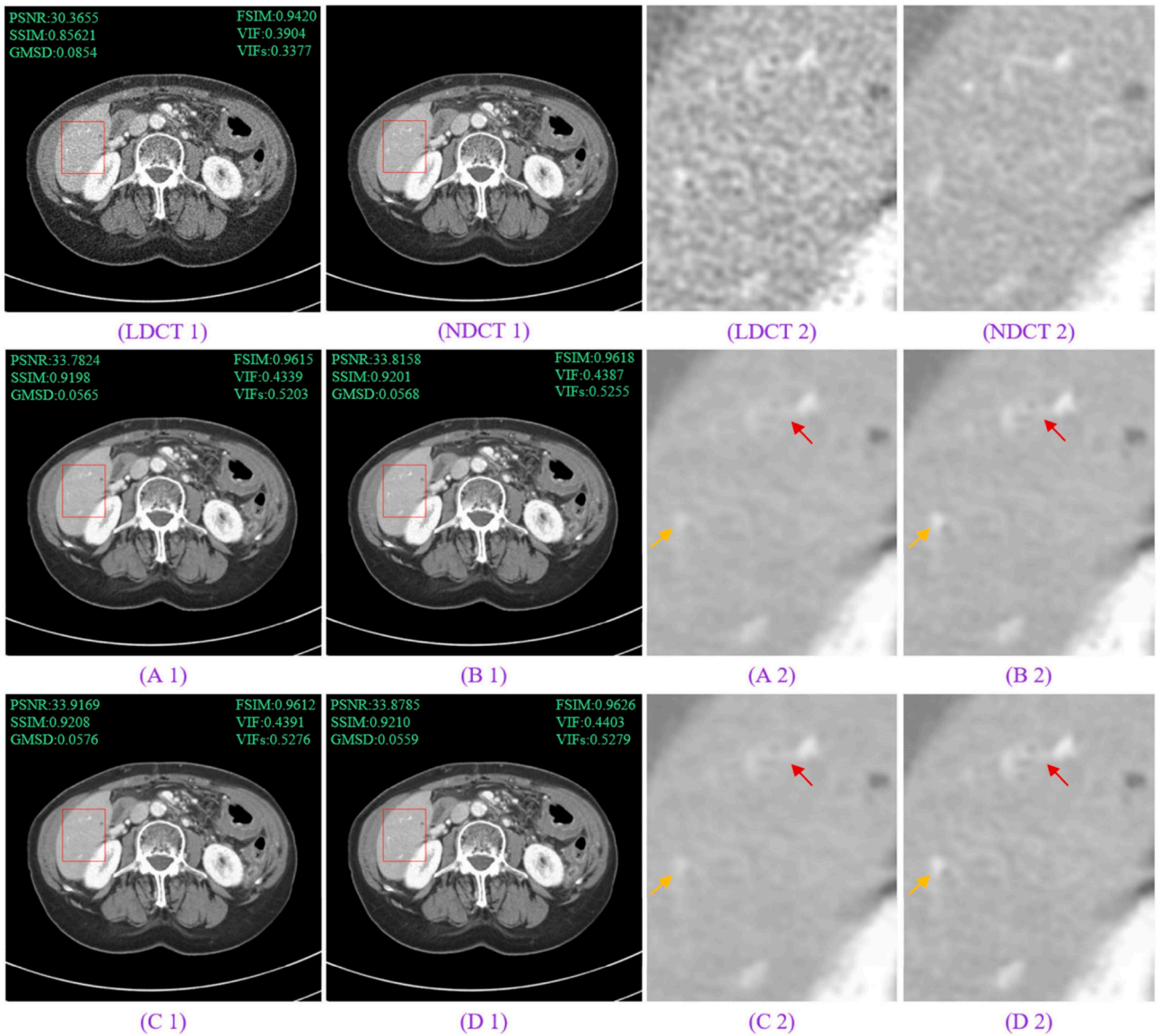


Fig. 8. The denoised results (Case 5 slice) of the ablation experiments on Mayo dataset [59].

computational performance [76]. Following, a computational framework use ResNet-50 CNN to classify mammogram images for diagnosing breast cancer. The transfer learning is also utilized from the pertained ResNet-50 CNN on ImageNet [77]. Dey et al. [78] designed copula-based model integrates consisting three best performing CNN architectures (DenseNet-161/201, ResNet-101/34, InceptionNet-V3). The novel framework showed a gratifying performance in malignancy prediction.

In order to solve the problem that overall accuracy in predicting lung tumor is low, Pandit et al. [79] devised a method with the auto encoder system to reduce processing time along with increase the overall prediction accuracy in pooling layer of CNN. The new model is used for reducing error, which improved accuracy for predicting lung nodule. In addition to that, tuberculosis is a chronic respiratory infectious disease with endangers human health. Xu et al. [80] considered a pulmonary tuberculosis classification algorithm that used CNN to classify chest X-ray images. The proposed method introduced coordinate attention mechanism into CNN, which can better classify and identify pulmonary tuberculosis images. In addition to lung, CNN is also widely used for heart disease diagnosis. Lin et al. [81] devised a multilayer one dimensional CNN-based classifier for automatic cardiomegaly level

screening. Two-round 1D convolutional process is used to identify normal condition and cardiomegaly levels. In Ref. [82], a fusion-based convolutional fuzzy neural network (F-CFNN) is designed for identifying and classifying CT images. The authors utilized a fuzzy neural network to provide robust classification results, and used two convolutional and two pooling layers to extract features. An optimized convolutional neural network model (ADECO-CNN) [83] is proposed for the COVID-19 detection in infected and not infected patients. An efficient deep convolutional generative adversarial network and convolutional neural network (DGCNN) is used in Ref. [84] for coronavirus disease 2019 (COVID-19) diagnosis, in which one generated fake images and the other differentiated between them. Alshudukhi et al. [85] illustrated an automatic registration algorithm to identify the translation, scale, and rotation for chest X-rays pneumonia images. Palomo et al. [86] applied a computer-aided diagnosis (CAD) system based on CNN to detect pneumonia for chest X-ray images. The experimental results indicated that the CAD system obtained significant performance. The authors in Ref. [87] trained the GoogleNet and ConvNet architecture to classify 425 CT-coronavirus images by using the COVID-CT dataset.

A lightweight CNN is formulated by Ahmad et al. [88] for extracting the liver region. It was demonstrated that the proposed method is

performance on three benchmark datasets including MICCAI SLiver'07, LiTS17, and 3Dircadb01. In Ref. [89], 3D networks used for 3D volume segmentation is introduced in addition to developing 3D networks. The proposed method significantly saved computational and processing time for developing 3D networks, and achieved high accuracy for 3D medical image segmentation. Chen et al. [90] applied convolutional neural network technique to classify myocardial perfusion images for coronary heart diseases. Therefore, the authors developed data-driven pre-processing to extract the area of interest. The prototype system can diagnose coronary heart diseases accurately in practice. Convolutional neural networks-based coronary motion artifact recognition and quantification is developed to assess the image quality and diagnostic reliability for coronary CT angiography images. Meanwhile, the authors developed and applied a coronary motion forward artifact model for CT data (CoMoFACT) for clinical cases with excellent image in Ref. [91].

Convolutional neural network is introduced for the automatic segmentation of gastric cancer images [92]. Moreover, the authors also designed a reiterative learning framework to train a simple network. The learning framework can achieve superior performance without pertaining or further manual annotation. In Ref. [93], an automatic gastric cancer segmentation with Deeplab v3+neural network is developed to improve the efficiency of cancerous regions and gastric cancer pathological slice image recognition. The results concluded that the novel method achieved good experimental results. Nagao et al. [94] devised ResNet 50, 3 independent artificial intelligence (AI) systems by using transfer learning and convolutional neural network architecture. A novel stepwise fine-tuning-based CNN scheme is introduced for gastric pathology image classification, in which a new type of target-correlative intermediate datasets is established. The numerical results indicated that the proposed scheme obtained the superiority and feasibility of for boosting the classification performance in Ref. [95]. A transfer learning framework by fine-tuning pre-trained CNN is developed in Ref. [96]. The authors utilized CNN to classify gastric images into three classes, such as chronic gastritis, low grade neoplasm, and early gastric cancer. Lui et al. [97] presented an AI image classifier with five convolutional neural network and three fully connected layers. The proposed classifier can accurately predict neoplastic component of gastric lesions. Skin cancer is one of the most severe diseases, convolutional neural network is among the main algorithm for cancer diagnosis. Salih et al. [98] presented a new CNN model with genetic algorithm to automatically configure a convolutional neural network model for early skin lesion diagnosis. Four public datasets are used to train and detect abnormalities for skin lesion features in different orientations. The model achieved the best scores than other existing methods. Villa-Pulgarin et al. [99] proposed a framework based on CNN and TL to analyze the classification performance in different skin lesions. The research results showed that the proposed methodology represented a helpful tool for skin cancer disease. Automated classification of skin lesions between melanoma and benign lesions is a challenging task. A new residual deep convolutional neural network (RDCNN) is introduced by Hosny et al. [100] for skin lesions diagnosis. The performance of the proposed RDCNN is measured by three different experiments. Soon afterwards, Sharma et al. [101] formulated a novel method, cascaded ensembling of convolutional neural network and handcrafted features based deep neural network, for dermatologist-level classification of skin cancer. The accuracy of convolutional neural network model is improved to 98.3 % from 85.3 %.

A method called transformed deep convolution network (TDCN) [102] is designed by using a novel deep learning architecture for multi-modal vertebra recognition. The proposed new architecture can automatically rectify the pose of vertebra. Hetherington et al. [34] introduced a system utilizing a deep CNN for classifying transverse images of the lower spine, which implemented several existing CNN architectures using TL. The numerical results indicated that the proposed novel system successfully identified lumbar vertebral levels.

From the above papers reviewed in this section, it is evident that

CNN has pervaded every aspect of medical image recognition. Alzheimer dementia diagnosis, brain multiple sclerosis lesions, brain tumor classification, retinal fundus images segmentation, hepatobiliary tumors diagnose, breast cancer detection, lung cancer classification, gastric cancer recognition, and skin cancer diagnosis, were the most studied applications. Based on the above surveyed articles, we found that most of the strategies are based on several common machines deep CNN methods, such as deep adversarial CNN, CNN based on transfer learning, CNN based on genetic algorithm, multilayered CNN, and CNN based on U-net architecture. It can be seen that the end-to-end deep trained CNN have become the preferred approach, challenging the accuracy of human experts for medical imaging recognition. Recognition in medical imaging has seen a huge influx of CNN related methods. A large diversity of deep architectures is covered. Custom architectures have been created to directly target the segmentation task and promising results have been obtained.

Although CNN are now clearly the top performers in most medical image recognition competitions, we can draw one striking conclusion that the architecture is not the most important determinant in getting a good solution. For example, many researchers used the same architectures and networks in challenges like the kaggle diabetic retinopathy challenge, that but have widely varying results. Expert knowledge can provide advantages that go beyond adding more layers to CNN, which is a key aspect that is often overlooked is that.

In addition, it can be seen from the above literature that CNN require under preferable circumstances a large amount of data for training, which led to the data scarcity problem. Particularly, the cost of expert-annotated data sets and the limited size of medical cohorts are some well-known challenges. Label noise can be a significant limiting factor in developing algorithms when data is annotated by domain expert. Whereas, the labeling of medical images is typically relatively low. For example, a widely used dataset for evaluating image analysis algorithms to detect nodules in lung CT is the LIDC-IDRI dataset. Pulmonary nodules were annotated by four radiologists independently in this dataset. How to deal with noise and uncertainty for training a CNN system on such data needs to be carefully considered. One could think of solutions like incorporating labeling uncertainty directly in the loss function, but this is still an open challenge.

Another data-related challenge is class imbalance. Images for the abnormal class might be challenging to find in medical imaging. To give an example, vast databases of mammograms are normal and do not contain any suspicious lesions for breast cancer screening programs. This is often not cancerous when a mammogram does contain a suspicious lesion. Even most cancerous lesions will not lead to the death of a patient. Designing deep learning systems with data augmentation strategies that are adept at combating this class imbalance is another important area of research.

3.2. Medical image identification used in deep learning

3.2.1. Deep learning

Deep learning (DL) [103] is a new research direction in the field of machine learning, which is introduced into machine learning to make it closer to the original goal [104,105]. The inherent rules and representation levels of sample data is learned by using deep learning, which is a complex machine learning algorithm. Deep learning is representation-based learning methods which use for classification purposes [106]. Deep learning refers to large neural networks where indicated a number of layers. In addition, deep learning provided automatic feature extraction for feature learning where can discover favorable patterns from available large amount of data. Deep learning is popular for mainly three reasons: 1) powerful central processing unit and high-performance computing devices, 2) large volume of data serves deep learning algorithms, and 3) creative algorithms for neural networks work [107].

Deep learning has brought revolutionary changes duo to effectively

train large neural networks. Different types of deep learning methods are applied in different areas, such as CNN, long short-term memory (LSTM), recurrent neural network (RNN), and generative adversarial networks (GAN) [107]. Deep neural networks are classified into two networks (shallow and deep), in which shallow network is generally with 1 or 2 hidden layers, and deep network with 3 or more hidden layers. The Deep learning mainly consists of three layers, such as an input layer or one to many convolutional 2-D layers (Conv2DL), pooling layers (PL), and fully connected layers (FCL). The functioning is as follows:

- 1) Input layer or convolutional 2-D layers: this layer consists of 2-D kernels move mainly input image. A 2-D filter consists of $n \times n$ dimension matrix, which is convolved with input images. The 2-D convolutional operation is distanced as shown in Eq. (7).

$$W[i,j] = \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} U[p,q]V[i-p,j-q] \quad (7)$$

where W expressed the output image, V is the input image matrix, and U is the kernel matrix. image matrices is processed through the indices i and j , and kernel matrices is processed through p and q . X_{in} is width for an input chest CT scan image. height Y_{in} with C_{in} number of channels. X_{op} and Y_{op} represented the output image width and height, respectively. X_{op} and Y_{op} are distanced, as shown in Eqs. (8) and (9).

$$X_{op} = \left\lfloor \frac{X_{in} - n + 2p}{s} \right\rfloor + 1 \quad (8)$$

$$Y_{op} = \left\lfloor \frac{Y_{in} - n + 2p}{s} \right\rfloor + 1 \quad (9)$$

where n expressed the matrix of dimension, s represented the stride.

- 2) Pooling layers (PL): This layer is known as the sub-sampling or down-sampling layer. The number of the Conv2DL neurons is decreased by the pooling operation, which prevented computational complexity for over fitting and minimize. The average pooling function and the max-pooling function are used to speed up calculations.
- 3) Fully connected layers (FCL): Fully-connected express each the upper layer neuron is connected to every next layer neuron. Fully connected layers similarly to a standard multi-layer perceptron neural network, in which each input connected to each output by a learning

weight. The 2D output feature map is transform into 1D in FCL. A softmax function is utilized with each value ranges between 0 and 1 aims to assign the output value from the last FCL to the desired class probability. The layered architecture of the deep learning is shown in Fig. 9.

3.2.2. Deep learning applied on medical images

Medical images are becoming more and more important in clinical diagnosis when the development of medical imaging technology. Deep learning is widely used in medical images processing [108]. We summarize the recent literature using deep learning technology for five parts of the human body. Among them, 7 papers were published for the breast and cervical parts of the human body, accounting for 26 %. 4 papers were published for the head (eye, nose, and thyroid) parts of the human body, accounting for 15 %. 5 papers were published for the brain and cranial nerve parts of the human body, accounting for 18 %. 8 papers were published for the chest (stomach, lung cancer, and lung nodule) parts of the human body, accounting for 30 %. 3 papers were published for the bone (Hip, pediatric musculoskeletal) parts of the human body, accounting for 11 %, as shown in Fig. 10. Review of head, bone, chest, brain, and breast pathologies detection using deep learning are shown in Tables 7–10, respectively.

Disease grading and fine-grained lesion segmentation are two important research areas for medical image analysis, in which pixel-wise data annotation for medical images is highly time-consuming. Pixel-wise data annotation for medical images is highly time-consuming. Zhou et al. [109] developed a collaborative learning method by semi-supervised learning with an attention mechanism, which improved the performance for disease grading and lesion segmentation. The severity classification accuracy is improved by designing a lesion attentive disease grading model. At the same time, the lesion maps are refined by using class-specific information and the segmentation model with semi-supervised manner. The effectiveness of proposed method is validated on a representative medical problem called diabetic retinopathy (DR). Hemelings et al. [110] developed a deep learning-based classifier for glaucoma diagnosis which pooled 8433 retrospectively fundus images. This study achieved automated glaucoma referral through combination of careful data augmentation, uncertainty sampling, and transfer learning. The experiments showed that deep learning can achieve excellent performance in glaucoma classification. Guo et al. [111] developed a hybrid intelligence-driven medical image recognition (HIM-MI) framework combined deep learning with the conventional

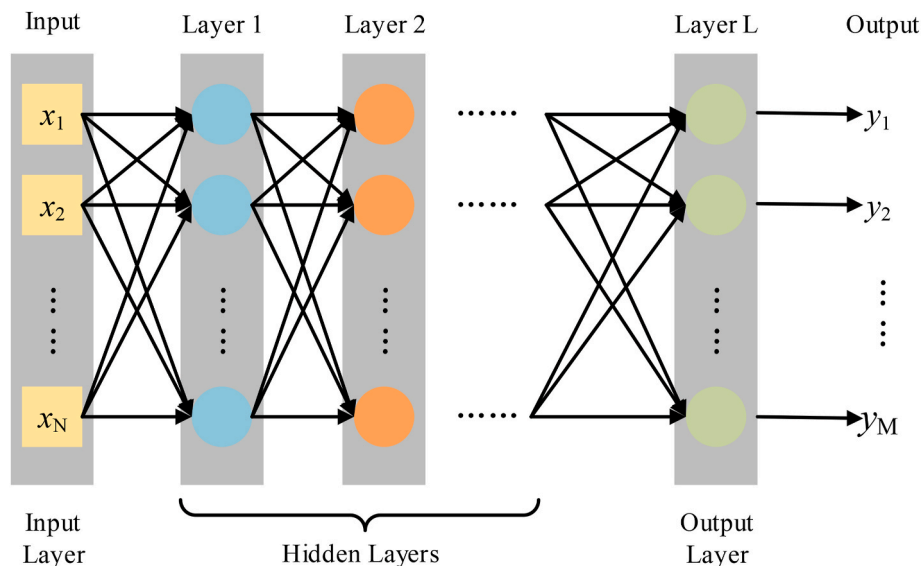


Fig. 9. Layered architecture of the deep learning.

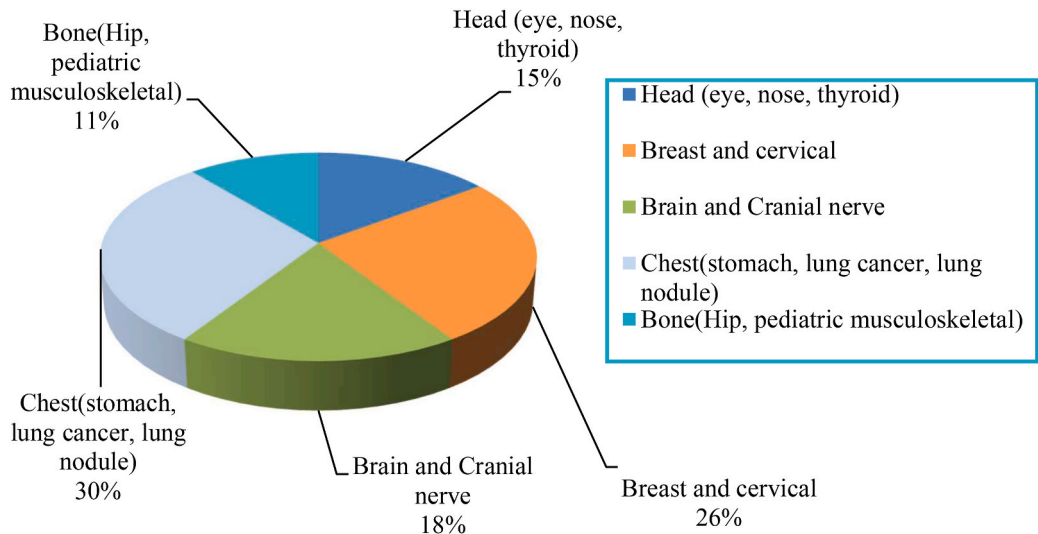


Fig. 10. Classification and proportion of medical image identification used in deep learning.

Table 7
Review of head pathologies detection using deep learning.

Reference	Year	Pathology	Modality	Proposed approach
Zhou et al. [109]	2019	Eye	Retinopathy classification	Collaborative learning
Hemelings et al. [110]	2020	Eye	Glaucoma diagnosis	Deep learning and transfer learning
Guo et al. [111]	2021	Eye	Retinal image recognition	Hybrid intelligence-driven framework
Xu et al. [112]	2022	Nasopharynx	Nasopharynx cancer recognition	Deep Learning
Zhang et al. [113]	2018	Thyroid	Human thyroid recognition	Generative adversarial networks
Khairandish et al. [114]	2022	Brain	Brain tumor detection	Threshold segmentation
Thanellias et al. [115]	2023	Brain	Subarachnoid hemorrhage recognition	Deep learning
Gon et al. [116]	2023	Brain	Restricting features classification	Spatial attention recurrent network
Dourado et al. [117]	2020	Brain	Lung nodule classification	Online computational intelligence learning
Fang et al. [118]	2023	Cranial nerve	Major depressive identification	UFA-Net

machine learning (CML), in which CNN is utilized to extract deep and abstract features for initial images. The CML-based technique is employed to construct a strong classifier that output recognition results. The proposed method is compared with four typical DML-based recognition models. The study results revealed that the HIM-MI improved recognition accuracy about two to three percent, which showed better recognition performance than four comparison methods.

Nasopharyngeal carcinoma (NPC) is a malignant tumor arising which is extremely difficult for the early diagnosis, because most patients are asymptomatic. Based on this, Xu et al. [112] introduced a deep-learning-based automatic diagnosis system, Siamese deep convolutional neural network (S-DCNN), to identify NPC from nasopharynx images. The experimental results indicated that CAD system simultaneously using two kinds of images, which achieved high patient accuracy and image accuracy. Zhang et al. [113] designed a new image

Table 8
Review of chest pathologies detection using deep learning.

Reference	Year	Pathology	Modality	Proposed approach
Kollias et al. [119]	2023	Chest	Lung cancer diagnosis	Recurrent neural network
Faes et al. [120]	2019	Chest	Multiple classification tasks	Automated deep learning platforms
Brehar et al. [121]	2020	Chest	Cirrhotic liver areas classification	Deep-learning-based solution
Wang et al. [122]	2020	Chest	Lung nodule detection	Semi-supervised learning framework
Heidari et al. [123]	2020	Chest	Pneumonia detect and classify	Deep learning model
Huang et al. [124]	2021	Chest	Chest segmentation	Multimodal GLORIA
Gupta et al. [125]	2023	Chest	COVID-19 automated screening	End-to-end framework with deep learning
Lee et al. [126]	2019	Stomach	Gastric cancer recognition	Deep neural network and transfer-learning

Table 9
Review of bone pathologies detection using deep learning.

Reference	Year	Pathology	Modality	Proposed approach
Cheng et al. [127]	2019	Hip	Hip fractures detection	Deep convolutional neural network
Yi et al. [128]	2019	musculoskeletal	Pediatric musculoskeletal classification	Deep convolutional neural networks
Yan et al. [129]	2016	Bodypart	Bodypart recognition of CT slices	Multistage deep learning framework

recognition approach with an adversarial learning-based, which is using to synthesize medical images. A generative adversarial network (GANs) is consisted for a generative network and a discriminative network in this research, which is applied for developing a medical image synthesis model. Three GANs, such as boundary equilibrium GANs (BEGANs), boundary equilibrium GANs (BEGANs) and deep convolutional GANs (DCGANs) are compared and implemented to synthesize medical images. The effectiveness for the generative network, trained to detect the synthetic images from real images, is validated by a discriminative network.

Table 10
Review of breast and cervical lesions pathologies detection using deep learning.

Reference	Year	Pathology	Modality	Proposed approach
Perek et al. [130]	2019	Breast	Breast mass distinguish	Deep learning with decision support system
Ya et al. [131]	2019	Breast	Breast lesions semantic segmentation	End-to-end deep learning
Shen et al. [132]	2019	Breast	Breast cancer detection	End-to-end deep learning
Zhang et al. [133]	2019	Skin	Melanoma diagnosis	Synergic deep learning model
Valkonen et al. [134]	2019	Breast	Breast cancer diagnostics	Deep learning
Clancy et al. [135]	2020	Breast	Breast cancer screening	Deep learning
Rahaman et al. [136]	2021	Cervical	Cervical cancer classification	Hybrid deep feature fusion

Hybrid method for brain MRI images [114] is investigated to detect and classify the tumor by using the BRATS database. In this research study, brain images are tried to classify as a benign and malignant tumor by using supervised hybrid support vector machine (SVM) and CNN techniques. The experimental results showed that the proposed hybrid model provides more effective and improvement techniques for classification compared with others. In Ref. [115], a deep learning algorithm identifying and localizing subarachnoid hemorrhage (SAH) on head CT scans is trained. Noncontract head CT images of patients for Helsinki university hospital with 90 head CT scans between 2012 and 2017 are used to identify and localize SAH. Performance of the trained algorithm is tested using data sets (137 SAH and 1,242 control cases). The results indicated that the propose method attained excellent performance. Gong et al. [116] introduced a new spatial attention recurrent network (SARN) framework based on Bayesian feature selection, in which discovered interpretable regional brain biomarkers. Features of these biomarkers are utilized to identify the addiction-related brain network patterns dynamically. Experiments are designed to investigate the superiority of the encoder and feature selector in the proposed framework. The better results are obtained by using the proposed graph representations for classification.

A new online computational intelligence framework based on deep learning tools is developed, which use with the internet of health things (IoHT) devices. The framework is validated on three medical databases (cerebral vascular accident images, lung nodule images, and skin images). The experimental results indicated that the proposed method attained excellent performance [117]. Fang et al. [118] proposed a discrepancy-based unsupervised cross-domain functional magnetic resonance imaging (UFA-Net) for automated major depressive disorder (MDD) identification. A maximum mean discrepancy constrained module is designed for feature alignment between unlabeled target domains and labeled source. The experimentation is conducted on 681 subjects with rs-fMRI data. The results indicated that the proposed method facilitated helps localize disease-associated functional connectivity abnormalities.

The authors in Ref. [119] developed a new large database of chest 3-D CT scans which is consisted by different numbers of CT scan slices in various context. In the meantime, a deep neural network based on convolutional neural network and recurrent neural network (CNN-RNN) architecture is designed to use for COVID-19 diagnosis on this database. A new large database of chest 3-D CT scans are introduced to use in the competition part of the ECCV 2022 AIMIA workshop, targeting COVID-19 detection and targeting COVID-19 severity detection. The datasets are used for validation of ML models and evaluation of the developed models. The experimental results showed that the performance of the proposed method outperforms the baseline scheme.

In [120], a deep learning architecture with neural architecture

search framework is studied to classify common diseases. In the framework, five publicly available open-source datasets (retinal fundus images, optical coherence tomography, images of skin lesions, adult chest X-ray and national institute of health (NIH)) are used to evaluate the diagnostic properties of the models. At the same time, the Edinburgh Dermofit library dataset is used for external validation. The experiments showed that diagnostic properties and discriminative performance were high in the binary classification tasks. Brehar et al. [121] proposed a deep-learning-based solution to differentiate the cirrhotic liver tissue areas. Image classification CNN-based is considered in the proposed deep-learning model, which is suitable for highly textured ultrasound images of liver. Other image classification CNN-based methods are compared with the proposed method on relevant textural features extracted. The experiments showed that the efficiency of the proposed solution is proved.

A semi-supervised learning framework named FocalMix is established by Wang et al. [122]. A raw medical image without annotation is utilized to improve the performance of supervised lesion detection models. Extensive experiments are carried out on two datasets lung nodule analysis (LUNA16) and national lung screening trial (NLST). The performance results showed that the FocalMix achieved a substantial improvement over state-of-the-art supervised learning approaches. Heidari et al. [123] considered a new CAD based on several deep learning models to detect and classify chest COVID-19 X-ray images. Two image preprocessing steps are applied for removing the majority of diaphragm regions. A histogram equalization algorithm and a bilateral low-pass filter are used to process the original image which is fed into three input channels based on transfer learning and CNN. Chest X-ray images are classified into 3 classes, such as Fig. 11. A publicly available dataset involving 8474 chest X-ray images is used for building and testing the CNN model. The experimental results showed that the propose method improved accuracy for detecting COVID-19 infected pneumonia.

An multimodal global-local representation learning framework based on deep learning for learning global and local representations is proposed in Ref. [124], in which the representations are learned by using attention-weighted image sub-regions and words. The analysis of the computational results indicated that the proposed method demonstrated high-performance and label-efficiency for classification and segmentation on different datasets. In the last few years, coronaviruses are a vast group of viruses which induced a variety of infections for serious respiratory problems. Gupta et al. [125] presented an end-to-end framework for the screening of the most transmissible disease (COVID-19) by using CT-scan images. The highest classification accuracy of 98.91 % is achieved by using ten-fold holdout validation method for transfer-learned DarkNet19, which showed the effectiveness of the presented method. Lee et al. [126] investigated a novel approach congaing deep neural network and transfer-learning to classify benign ulcer and cancer for gastrointestinal endoscopic color images. The approach is tested on a dataset comprised of 367 cancers, 200 normal, and 220 ulcer cases. The results indicated that the approach haven significant performance differences compared with others for clinical environments.

Cheng et al. [127] designed a deep convolutional neural network (DCNN) for the localization and detection of hip fractures on plain frontal pelvic radiographs (PXR), which improved the efficiency and accuracy of hip fracture diagnosis. A DCNN was pertained retrained using 3605 PXR between August 2008 and December 2016. The performance was evaluated on 100 independent PXR under the receiver operating characteristic curve (AUC). The algorithm achieved an accuracy of 91 %, a sensitivity of 98 %, a false-negative rate of 2 %, and an AUC of 0.98 for identifying hip fractures. The proposed DCNN method detected a low false-negative rate for hip fractures on PXR and had high accuracy for localizing fracture lesions. Grad-CAM-assisted image identification of hip fractures, as show in Fig. 12.

A deep convolutional neural networks (DCNN) [128] is established for pediatric musculoskeletal radiographs classification. DCNN are

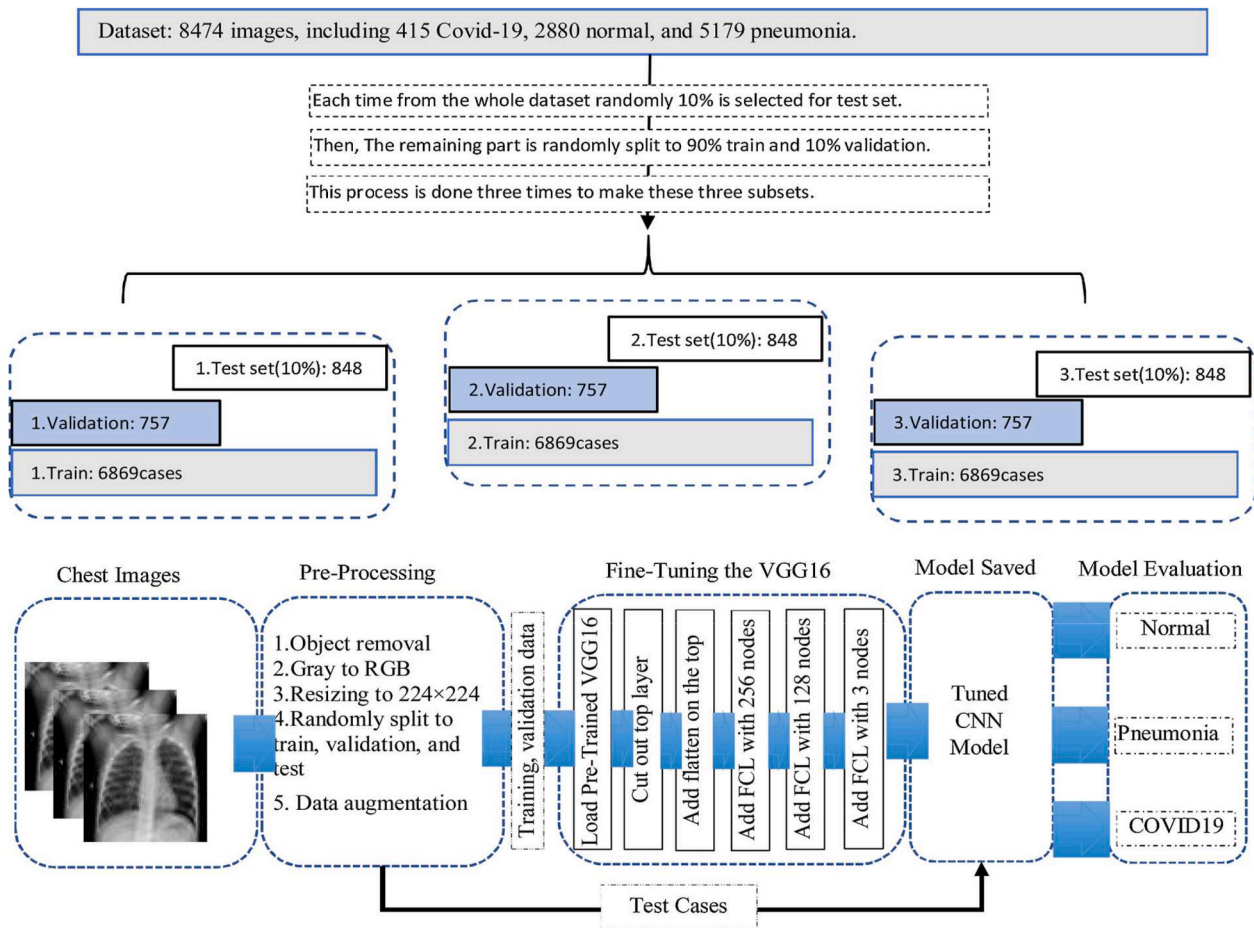


Fig. 11. Schematic representing training and validation phase of the proposed scheme [123].

trained on database of 250 pediatric bone radiographs with 50 each of the elbow, shoulder, knee, pelvis, and hand. The experimental results indicated that the proposed DCNN have good-to-perfect accuracy and superhuman speeds for pediatric musculoskeletal radiographs classification. Yan et al. [129] investigated a multistage deep learning framework to classify and recognize image for body part recognition. Two tasks are achieved by the two stages of learning scheme in this study. The key hallmark of the proposed method is that the discriminative and non-informative local patches are automatically discovered through multi-instance deep learning. The new network is validated on a large scale CT dataset and a synthetic dataset. The analysis of the computational results indicated that better performances are achieved compare with the standard deep CNN.

Purpose contrast-enhanced spectral mammography (CESM) is a recently developed breast imaging technique. Because of the large lesion variability, differentiation between benign and malignant enhancement is not accurate by CESM. A deep learning-based decision support system is presented by Perek et al. [130]. Two analysis approaches (fine-tuning a pertained network and fully training a convolutional neural network) are compared to distinguish CESM breast mass between benign and malignant. Each classification network is improved as an additional input to the network by incorporating breast imaging reporting and data systems (BIRADS) textual features. Fivefold cross-validation on 129 randomly selected breast lesions annotated is evaluated. Ya et al. [131] developed end-to-end deep learning approaches based on fully convolutional networks (FCNs) for semantic segmentation of breast lesions. In the model, imageNet and transfer learning are used to retrain for overcoming the issue of data deficiency. The experimental results are evaluated on two datasets (a total of 113 malignant and 356 benign

lesions). The results showed that the performance of the proposed method outperforms for benign lesions.

In the same way, Shen et al. [132] proposed an “end-to-end” deep learning algorithm with detected breast cancer through screening mammograms. In proposed method, all convolutional network method attained excellent performance for classifying screening mammograms. Zhang et al. [133] considered a synergic deep learning (SDL) model based on multiple deep convolutional neural networks (DCNN) to solve significant intra-class variation and inter-class similarity problem. Learned image representation is concatenated as the input of a synergic network for each pair of DCNN. When a correct classification is made, a synergic error will be leaded by the other mistake DCNN. This structure has a fully connected structure to predict whether the pair of input images belongs to the same class. This model is trained end-to-end under the supervision of classification errors from each pair of DCNN.

In [134], Valkonen et al. developed a deep learning based digital mask for automated epithelial cell detection by using two staining with fluoro-chromogenic cytokeratin-Ki-67 double staining and sequential hematoxylin-IHC staining. Meanwhile, a partially pre-trained deep convolutional neural network was fine-tuned from 152 patient samples of invasive breast tumors. A good discrimination of epithelial cells was achieved, which is well in concordance with pathologists’ visual assessment. The results concluded that deep learning can be applied to detect carcinoma cells in breast cancer. Coincidentally, Clancy et al. [135] evaluated deep learning performance for distinguishing mammography with malignancy and normal (benign) findings. Six different convolutional neural network model structures with four different imaging datasets are assessed for 1303 breast cancer screening patients. The experimental results showed that the strategy had

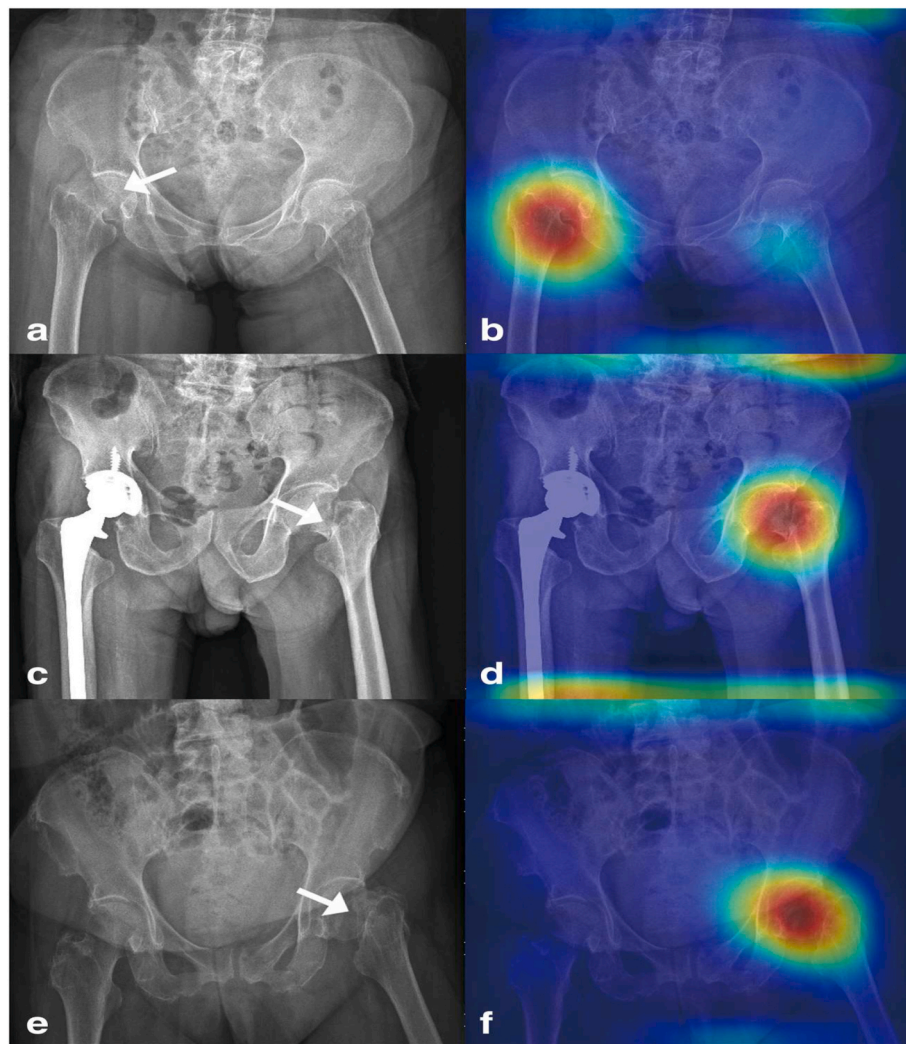


Fig. 12. Grad-CAM-assisted image identification of hip fractures [127].

significant performance compared with others.

In addition, deep learning is also used for cervical cancer identification. Cervical cancer is one of the most common fatal cancers among women. To improve the manual screening practice, a hybrid deep feature fusion (HDFF) technique is developed by Rahaman et al. [136] to solve the classification stage's optimality problem, which improved the performance for multiclass classification task with uneven distribution of data. The HDFF is tested on the publicly available SIPaKMeD dataset. The analysis of the computational results indicated that proposed method achieved a high accuracy and classification results.

We surveyed above articles on deep learning for medical image recognition. The results indicated an increased interest in the field. Retinal image recognition, tumor classification, lung nodule classification, gastric cancer recognition, hip fractures detection, and breast mass distinguish were the most studied applications. Therefore, based on the surveyed articles, we found that most of the strategies are based on several common deep learning methods, such as end-to-end deep learning, deep convolutional neural network, multistage deep learning framework, automated deep learning, hybrid intelligence-driven framework, generative adversarial networks.

The deep learning models introduced above provide useful theoretical support and practical experience for image identification. The image identification model based on deep learning has more obvious advantages compared with traditional image identification algorithms. The deep learning model learns the feature information of the source images

by training on a larger data set, and enhances the expression ability of the network for image feature information through continuous iterative process, but this also makes it more dependent on large data sets. However, deep learning still faces many problems in medical image identification. Comparing with other natural image datasets, the available datasets in the field of medical image is very limited, which makes the training of deep learning models difficult. Although traditional methods can be used to alleviate the problem of small sample datasets, it is still not the best way to address this issue. Therefore, it is very important for researchers in the field of deep learning and hospitals to cooperate to find more efficient methods of medical data annotation. Meanwhile, the generalization ability of deep learning is poor. A model can obtain good performance on the images collected by a specific device. When different devices are used, the performance will generally decline, which is also an urgent problem to be solved.

In addition to these, challenges in meaningful application of deep learning in object detection are thus mostly similar to those in object classification. Only few papers directly address issues specific to object detection, such as class imbalance, hard-negative mining, efficient pixel, voxel-wise processing of images. We expect that the application of multi-stream networks in a fully convolutional fashion will be given more emphasis in the near future.

3.3. Medical image identification used in machine learning

3.3.1. Machine learning

Machine learning (ML) [137] is an intelligent method that learns general rules from limited observational data and uses these rules to predict unknown data, which is a subset of AI [138–140]. The purpose of machine learning is to train models which are able to make good predictions about unknown data. Computers show automatically learns ability which improved without being explicitly programmed. Machine learning has been applied to many different fields, such as biomedical, finance, and manufacturing industry problems and so on [141–143]. ML method is generally divided into supervised learning, unsupervised learning, and reinforcement learning algorithms. Supervised learning includes the following algorithms: support vector machine, transfer learning, decision tree, rule-based learners, instance-based learning, artificial neural network, Bayesian neural network, native Bayesian, and genetic algorithm. Unsupervised learning includes the following algorithms: decision tree, rule-based learners, instance-based learning, artificial neural network, Bayesian neural network, native Bayesian, and genetic algorithm. Reinforcement learning includes the following algorithms: genetic algorithm, estimated values function, and simulated annealing. The classification of machine learning technique is shown in Fig. 13.

Classification, clustering, and regression are the main tasks for machine learning [144,145]. Classification and regression tasks are supervised learning which learn from a set of labeled data. Clustering is unsupervised learning that does not have labeled data. Characteristics of supervised and unsupervised learning aim to build a model with a set of labeled instances to predict the output of an unseen instance. In supervised learning, a model is presented with a dataset $D = \{x, y\}_{n=1}^N$ including input features x and label pairs y . This y can take several forms along with different the learning task. The y , a classification setting, is generally a scalar representing a class label. The y can be a vector of continuous variables in case of regression. The y can also be a multi-dimensional label image when a segmentation model is tried to learn. Supervised training searches the model parameter θ based on a loss function $L(y, \hat{y})$ with best predict. The $\hat{y}$ presents the output of the

model, which feed a data point x to the function $f(x, \theta)$.

In contrast, unsupervised learning algorithms perform unknown data learning in the absence of data labels and trained to find patterns. Examples of traditional unsupervised learning algorithms are principal component analysis and clustering methods. An example of traditional unsupervised learning algorithms consists of two types: clustering and principal component analysis. When there are many different loss functions, unsupervised training can be performed. When dealing with machine learning problems, it is common to have several steps, as shown in Fig. 14.

3.3.2. Machine learning applied on medical images

The application of machine learning in medical images is mainly distributed in brain, eye, lung cancer, shoulder, white blood cell, nervous, and chest regions. We summarized a part of the literature as a reference. In this section, the literature on head is the majority, accounting for 50 %. The literature on shoulder, head, shoulder, white blood cells, nervous and chest accounted for 10 %, respectively. Review of brain, eye, lung cancer, shoulder, white blood cell, nervous and chest regions using machine learning are shown in Fig. 15 and Table 11.

Alcaide-Leon et al. [146] used a texture-based ML for support vector machine classification with textural features of contrast-enhanced T1WI, which is no less than expert human evaluation to identify enhancing gliomas and primary central nervous system lymphoma. 35 adult patients with primary central nervous system and 71 adult patients with enhancing gliomas were included. The resulting volumes of interest were mined and subjected to a support vector machine-based protocol. The analysis of the computational results indicated that novel method achieved better performances than other. Castellazzi et al. [147] applied novel method combining with advanced MRI features to supportive in classifying vascular dementia (VD) from Alzheimer disease (AD), which is helpful to predict the prevalent disease in subjects with an unclear profile of vascular dementia or AD. The best feature pattern to classify VD from AD is fined utilizing diffusion tensor imaging (DTI) of 60 subjects (33 AD, 27 VD). The novel method attained excellent predictions performance compared with other algorithms.

An ensemble neural network based on Akanke information criterion

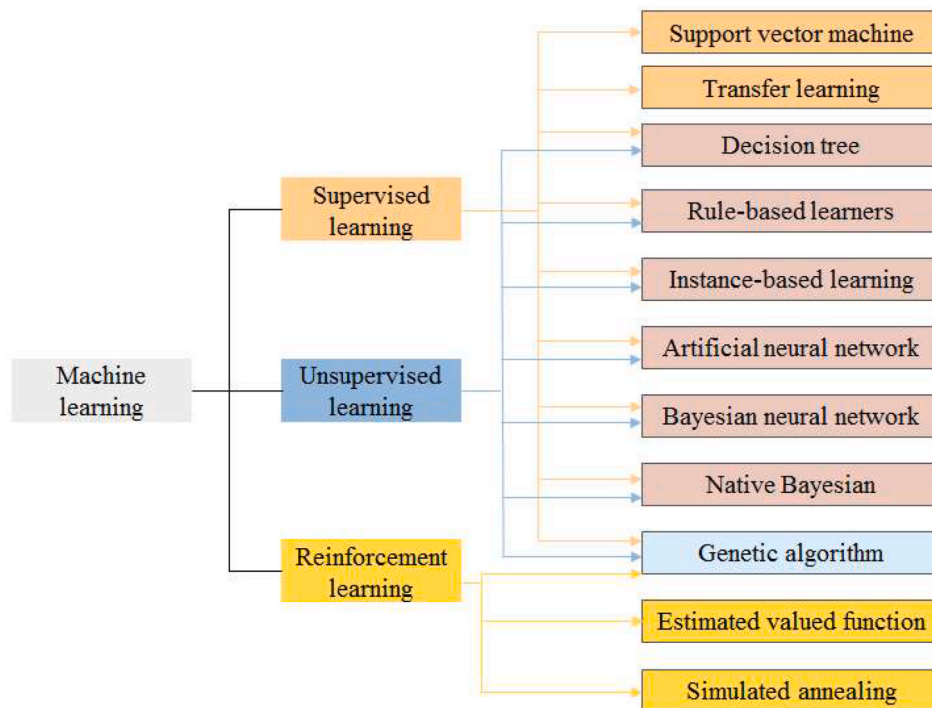


Fig. 13. Classification of machine learning technique.

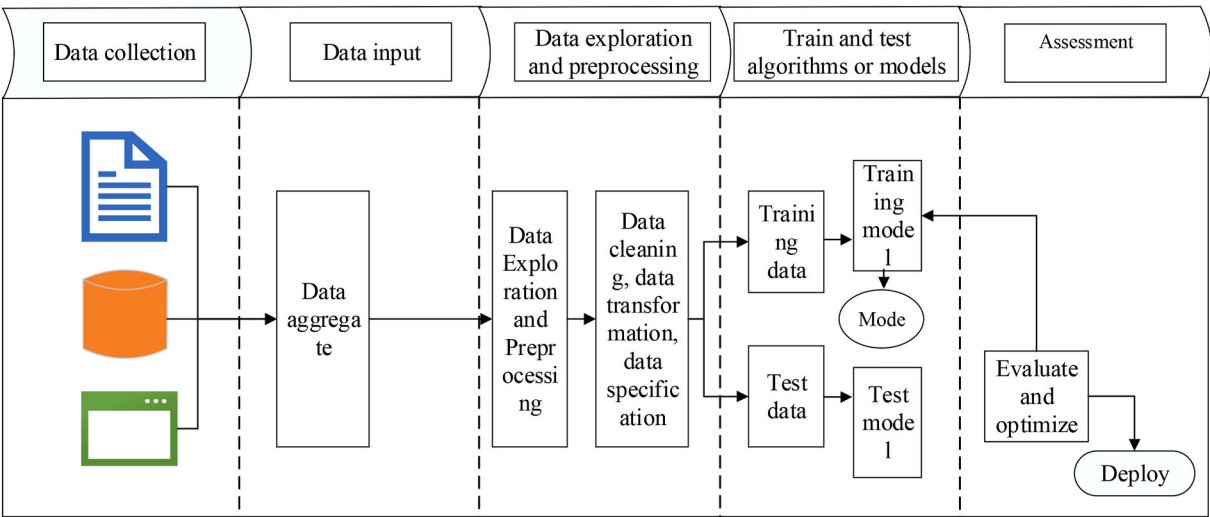


Fig. 14. Flow chart of machine learning steps.

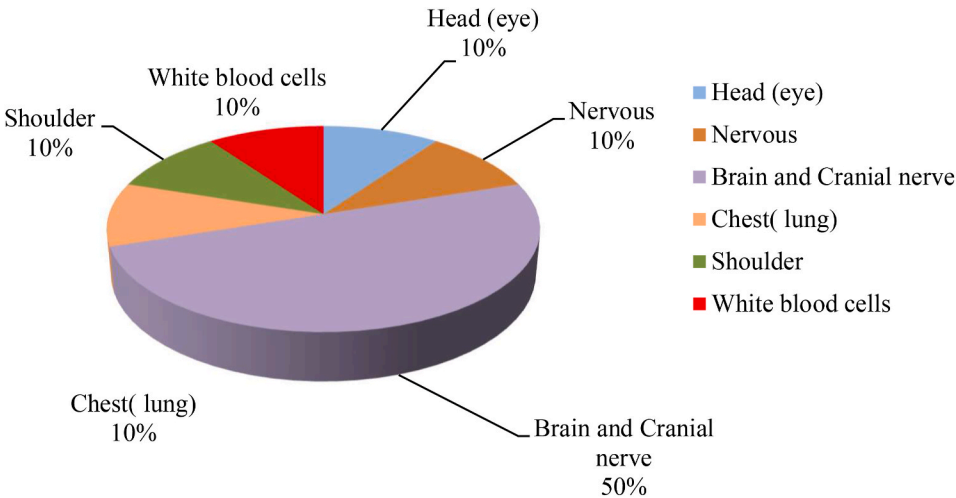


Fig. 15. Classification and proportion of medical image identification used in machine learning.

Table 11
Review of medical image identification using machine learning.

Reference	Year	Pathology	Modality	Proposed approach
Alcaide-Leon et al. [146]	2017	Brain	Enhancing glioma differentiation	Texture-based machine learning
Castellazzi et al. [147]	2020	Brain	Alzheimer dementia diagnosis	ML combining MRI features
Fooladi et al. [148]	2019	Brain	Multiple sclerosis lesions	Ensemble neural networks
Lee et al. [149]	2020	Brain	Estimate stroke onset	Automated machine learning
Stamile et al. [150]	2015	Brain	Multiple sclerosis classification	Complex network analysis
Geng et al. [151]	2023	Shoulder	Shoulder arthroplasty recognition	Machine Learning
He et al. [152]	2020	Lung	Circulating tumor cells recognition	Machine learning
Ismael et al. [153]	2020	blood cells	White blood cells recognition	Automatically identifies system
Tian et al. [154]	2021	Eye	Alzheimer's disease classification	Highly modular machine learning
Zhang et al. [155]	2016	Nervous	Multiple sclerosis detection	Machine learning

(ENN-AIC) is proposed in Ref. [148]. Patients with relapsing-remitting multiple sclerosis (RRMS) is classified by using quantitative magnetization transfer imaging (QMTI). The experimental results indicated that the ENN-AIC achieved 90 % accuracy, which indicated it is an effective neural network compared with multilayer perceptron (MLP) and radial basis function (RBF). Following [149], Lee et al. devised an automated machine learning system for identifying patients with acute ischemic stroke within 4.5 h of symptom onset, in which diffusion weighted imaging (DWI) and fluid-attenuated inversion recovery (FLAIR) images of consecutive patients with acute ischemic stroke within 24 h is analysed by applying automatic image processing approaches. The research results showed that machine learning techniques is useful for identifying therapy among patients with unclear stroke onset time.

Stamile et al. [150] introduced a complex network graph-based method to classify multiple sclerosis patients. SVM classification was executed utilizing weighted and unweight graphs. A new machine learning model is designed to identify shoulder implant type and manufacturers. The shoulder implants identified with high accuracy and speed by the new machine learning mode. The experimental results showed that the model performed with an overall accuracy of 93.9 % as shown in Fig. 16, which indicated greater performance [151]. He et al. [152] applied machine learning to recognize medical image for circulating tumor cells (CTC), which is used to monitor therapeutic efficacy

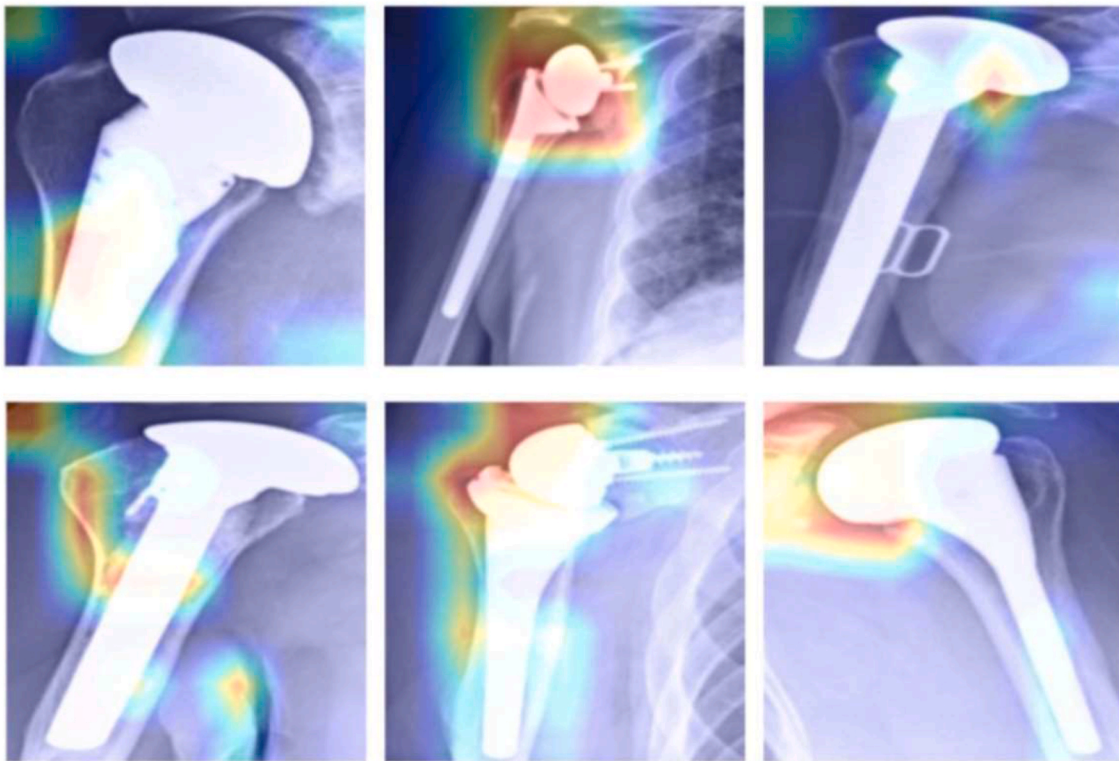


Fig. 16. Saliency mapping showing correct and incorrect identifications [151].

and tumor recurrence. The CTC test results contain 600 patients are collected to segment by image demising, edge detection, image filtering, contraction techniques and image expansion using python scheme. The specificity and sensitivity of recognition reached 91.3 % and 90.3, respectively.

In [153], Ismael et al. designed an automatically identifies system based on different ML algorithms for white blood cells (WBCs) in the medical images. The system is implemented in four steps: 1) segmentation of the cell images, 2) prepared its dataset for each segmented image, 3) extracting the textures and shapes from scanned image, 4) different machine learning algorithms. The simulation experimental results demonstrated that the method improved the important features, such as image intensity, shape, texture, and color. An alternative for conducting screenings [154] for Alzheimer's disease dementia patients is introduced, in which a highly modular machine learning technique is employed throughout the whole pipeline. The pipeline attained an average classification accuracy of 82.44 % by employing data from the UK Biobank. Zhang et al. [155] proposed a new system which included a multiple sclerosis (MS) detection method based on two-level stationary wavelet entropy (SWE) and k-nearest neighbors (kNN). In proposed method, the misclassification costs are adjusted to alleviate the effect for unbalanced class distribution. Meanwhile, two-level stationary wavelet entropy (SWE) is utilized to extract features from brain images. The analysis of the computational results indicated the effectiveness of proposed approach is superior to four state-of-the-art approaches.

We surveyed above articles on machine learning. The results indicated that machine learning technologies have attracted a remarkable level of attention from researchers in different fields. Alzheimer dementia diagnosis, brain multiple sclerosis lesions, tumor classification, and tumor segmentation were the most studied applications. Based on the surveyed articles, we found that most of the strategies are several common machine learning methods, such as automated machine learning, complex network, artificial neural network, and support vector machine.

Most of above machine learning technologies work under the

common assumption that the training data and the test data have identical feature spaces with underlying distribution. Once the feature space of the test data changes, the prediction models must be rebuilt and retrained from newly collected training data. It is very expensive and sometimes not practically possible. Similarly, it is nearly impossible to establish a learning-based model for very few labeled data available for supervised learning when learning-based models need adequate labeled data for training. Semi-supervised learning can partly solve this problem by relaxing the need of mass labeled data.

Although machine learning approaches are increasingly successful in medical image classification and image-based diagnosis, many practical challenges still need to be addressed to unlock their full potential. These challenges include that how to improve access to data, how to train strong models on little data, how to best make use of the image structure and specific properties of medical imaging data in designing our models, and how to apply these results in clinical practice.

3.4. Medical image identification used in transfer learning

3.4.1. Transfer learning

Transfer learning [156] stems from cognitive research, is a technique utilizing external expertise to benefit the learning process of the target domain, which becomes a crucial topic in reinforcement learning. TL improved performances of a new task by using an idea that knowledge is transferred across related tasks. Transfer learning is essentially about solving the perceived requirement deep network training with large data sets, in which two transfer learning strategies are identified. The first strategy is that a pre-trained network is used as a feature extractor. The second strategy is that a pre-trained network is fine-tuned on medical data. The study of transfer learning is motivated to solve new problems faster or with better solutions by applying knowledge learned previously. The difference between traditional machine learning and transfer learning is that traditional machine learning techniques try to learn each task from scratch, while transfers learning try to transfer the knowledge from some previous tasks when the latter has fewer high-quality training

data. Different learning processes between transfer learning and traditional machine learning is shown in Fig. 17. Transfer learning has attracted more and more attention since 1995, and they have different names: knowledge transfer, inductive transfer, learning to learn, multi-task learning, knowledge consolidation, context sensitive learning, incremental/cumulative learning, and meta learning.

The unified definition of transfer learning is given as follows [157]: given a learning task T_S and T_T , source domain D_S and target domain D_T , and a target predictive function $f_T(\cdot)$. Transfer learning can help to improve the learning of $f_T(\cdot)$ in target domain D_T utilizing source domain D_S and learning task T_S , where $T_S \neq T_T$, or $D_S \neq D_T$. A domain is pair $D = \{x, P(X)\}$, the condition $D_S \neq D_T$ implies that $x_S \neq x_T$ or $P_S(X) \neq P_T(X)$. Here is an example of document classification, this means that between a target document set and a source document set, either their marginal distributions are different between the two sets, or the term features are different. A task is defined as a pair $T = \{y, P(Y|X)\}$. The condition $T_S \neq T_T$ implies that either $P(Y_S|X_S) \neq P(Y_T|X_T)$ or $y_S \neq y_T$. The learning problem becomes a traditional machine learning problem when $D_S = D_T$ and $T_S = T_T$.

3.4.2. Transfer learning applied on medical images

In this section, the transfer learning applied to medical images is mainly distributed in brain, eye, lung, chest, breast, prostate, retinal, abdominal, and cardiovascular. We summarized a part of the literature as a reference. The literature on chest is the majority, accounting for 29 %. The literature on breast accounted for 22 %. The literature on brain, eye, lung, prostate, retinal, abdominal, and cardiovascular accounted for 7 %, respectively. Review of brain, eye, lung, chest, breast, prostate, retinal, abdominal, and cardiovascular using transfer learning are shown in Fig. 18 and Table 12.

Yu et al. [158] applied transfer learning to diagnose mammographic breast lesion. Meanwhile, deep neural network (DNN) models with large-scale natural images are employed. The analysis of the computational results indicated that involved models including transferred DNN achieved the best performance, which improved the mammographic breast cancer diagnosis. A multistage transfer learning (MSTL) algorithm utilizing three pre-trained models (EfficientNetB2, ResNet50, and InceptionV3) is designed in Ref. [159]. The experimentation is conducted in 20,400 cancer cell images, 200 ultrasound images from Mendeley, and 400 ultrasound images from the MT-Small-Dataset. The results indicated that the MSTL could significant improve the early diagnosis for breast cancer. Borkowski et al. [160] used transfer learning

effectively to classify the MRI breast images based on background parenchyma enhancement. The experimental results indicated that the convolutional part of the VGG16 network can serve as an effective feature extractor for breast MRI in the absence of not trained on medical images.

A transfer learning with deep convolutional neural networks is developed by the authors in Ref. [161] to classify abdominal ultrasound images. Two convolutional neural networks based on CaffeNet and VGGNet are connected, which previously trained on the 2012 large scale visual recognition challenge data set. The research results showed that transfer learning with convolutional neural networks achieved better performance for constructing abdominal ultrasound images effective classifiers. Authors in Ref. [162] introduced a TLradionics model combined the gray scale and electrogram ultrasound images to estimate accurate liver fibrosis grading. In the proposed model, electrogram modality (EM) and gray scale modality (GM) are analyzed by a deep convolutional neural network. The performance of the proposed method is validated by totally 466 patients undergoing partial keratectomy. Experimental results show the transfer learning modal combining EM and GM images obtained excellent performance. Mazo et al. [163] devised (CNN based transfer learning to automatically recognize and classify fundamental cardiovascular tissues and organs. We adapt four popular CNN architectures (ResNet, VGG19, VGG16, and inception) to solve this problem by using transfer learning. It was demonstrated that CNN based transfer learning hold great promise for improving tissue and organs classification. The proposed transfer learning approach for automatic classification of fundamental tissues was shown in Fig. 19. Karri et al. [164] investigated a transfer learning algorithm to identify retinal pathologies. In order to improve prediction capability, a pre-trained CNN is devised. The pre-trained convolutional neural network approach outperformed than conventional classical learning.

In [165], Meng et al. devised a two-stage transfer learning recognition model for medical images of COVID-19, in which the vision transformer (ViT) pertaining model is used to obtain generic features from massive heterogeneous data. The experimentation is conducted on a COVID-19 dataset. The results showed that the proposed method may greatly alleviate data scarcity problem. Da Nobrega et al. [166] investigated a deep transfer learning to improve the performance of lung nodule malignancy classification tasks. The experimental results indicated that deep transfer learning attained excellent performance for lung nodule CT images. Zhang et al. [167] introduced the classic LeNet-5 model with transfer learning to classify pulmonary nodules of thoracic

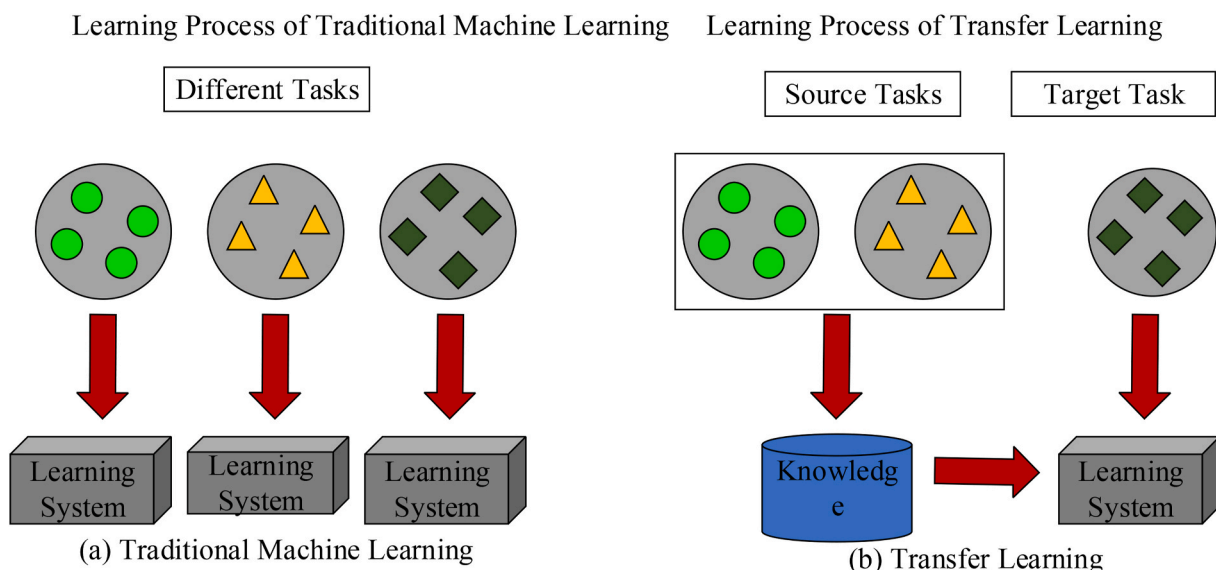


Fig. 17. Different learning processes between transfer learning and traditional machine learning.

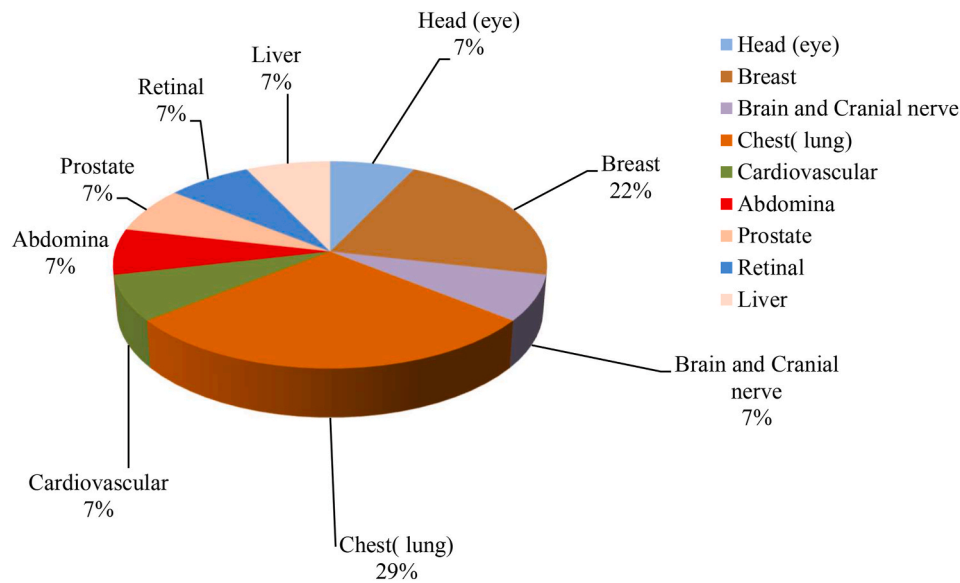


Fig. 18. Classification and proportion of medical image identification used in transfer learning.

Table 12
Review of medical image identification using transfer learning.

Reference	Year	Pathology	Modality	Proposed approach
Yu et al. [158]	2019	Breast	Mammographic breast diagnosis	Transfer learning
Ayana et al. [159]	2022	Breast	Breast cancer classification	Novel multistage transfer learning
Borkowski et al. [160]	2022	Breast	Breast MRI classification	Transfer learning
Cheng et al. [161]	2017	Abdominal	Abdominal ultrasounds classification	Transfer learning with deep CNN
Xue et al. [162]	2020	Liver	Liver fibrosis grading	Transfer learning
Mazo et al. [163]	2018	Cardiovascul	Cardiovascular tissues classification	CNN based transfer learning
Karri et al. [164]	2017	Retinal	Retinal pathologies identification	Transfer learning
Meng et al. [165]	2022	Chest	COVID-19recognition	Two-stage transfer learning
Da Nobrega et al. [166]	2022	Chest	Lung nodule classification	Transfer learning and CNN
Zhang et al. [167]	2019	Chest	Pulmonary nodules classification	Transfer Learning
Yuan et al. [168]	2019	Prostate	Prostate cancer classification	Multiparametric transfer learning
Nishio et al. [169]	2018	Lung	Lung nodule classification	Computer-aided diagnosis
Gómez-Valverde et al. [170]	2019	Eye	Glaucoma classification	CNN and transfer learning
Dawud et al. [171]	2019	Brain	Brain hemorrhage classification	Transfer learning

CT images which included pulmonary nodules and different malignancies of the malignant nodules. The robustness of the classification model is testified by implementing the 10-folder cross validation (CV). The experimental results indicated that transfer learning of the LeNet-5 achieved good classifying performance for pulmonary nodules images.

A novel multiparametric magnetic resonance transfer learning

method named MPTL for automatically staging prostate cancer is investigated in Ref. [168]. The novel method is established by deep convolutional neural network with three branch architectures. Pre-trained model is transferred to compute features based on multi-parametric MRI images (mp-MRI). The main result of this study showed that the proposed method can classify the cancer accurately and learn discriminative features for prostate images. Nishio et al. [169] designed a CAD method with transfer learning to classify benign nodule, primary lung cancer, and metastatic lung cancer. The experimentation is conducted in tomography images and clinical information of 1236 patients. The CAD was evaluated using the VGG-16 CNN with and without transfer learning. The results indicated that the proposed method improved the accuracy of lung nodule classification. Authors of [170] presented a method based on convolutional neural networks and transfer learning for glaucoma detection utilizing color fundus images. Three different data sets (RIM-ONE, two publicly available, and DRISHTI-GS) are used to assess the performance of the alternatives. It demonstrated a slight improvement in sensitivity and specificity with similar algorithms using the area under the receiver operating curve (AUC). Authors of [171] studied modified novel version of AlexN using SVM named AlexNet-SVM as a classifier instead of a three layer neural network for the brain CT images, which applied the same classification task into three deep networks. The study results showed that the proposed modified AlexNet-SVM can outperform convolutional neural network.

Above articles indicated an increased interest in the field in recent years. Tumor classification and tumor segmentation were the most studied applications. Therefore, based on the surveyed articles, it is found that most of the strategies pretrained a model or utilized a pre-trained model and subsequently finetuned certain parameters, or a separate model on the target domain data. Among these studies, a similar number of approaches either utilized pretrained or image Net-pretrained CNN on medical images. In classical machine learning problems, neural network have be demonstrated the superior performance by many studies compared to statistical methods. To address the problem in transfer learning, part of the literature presented above applied the main deep learning network techniques, such as deep neural network, multiple tasks neural network, and radial basis function neural network to transfer learning.

The reason why transfer learning achieved good performance in clinical diagnostics is that transfer learning solved the problem of insufficient training data. Training dataset with an expensive price is

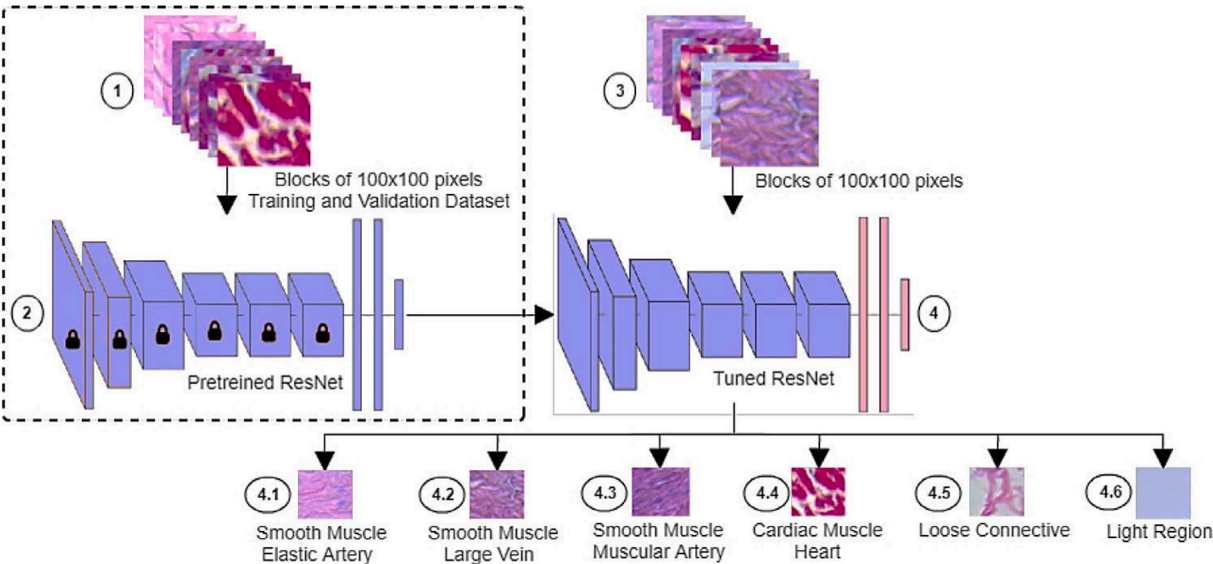


Fig. 19. Proposed transfer learning approach for automatic classification of fundamental tissues associated with an organ: (1) image blocks of 100×100 pixels belonging to training and validation dataset. (2) pre-trained network. (3) image blocks of 100×100 pixels. (4) fine-tuned model. (4.1), (4.2), (4.3), (4.4), (4.5) and (4.6) classified blocks [163].

very easy to get out of date, which cannot be effectively applied in the new tasks. However, transfer learning relaxed the hypothesis that the training data must be independent and identically distributed with the test data, which can significantly reduce the demand of training data and training time in the target domain. The need for transfer learning stemmed from the heterogeneity in large datasets combining MR images from scanner hardware and acquisition centers with different imaging protocols. Additionally, transfer learning can increase clinical applicability by simplifying the training in new environments.

Traditional machine learning techniques only try to learn each task from scratch, while transfer learning techniques try to transfer the knowledge from other tasks and domains to a target task. We can pave the way for construction of the learning-based model for the target

domain if we can transfer and exploit the knowledge from similar source domain, and as such we expect transfer learning to become even more prominent for this task in the near future.

3.5. Medical image identification used in other image processing technology

In addition to the above analysis of CNN, ML, DL, TL, and other methods, there are some special medical image technology have achieved excellent performance for image classification, recognition, and segmentation. In this section, we summarized the application of other image technology in medical image recognition. The other image processing technology in medical images is mainly distributed in brain,

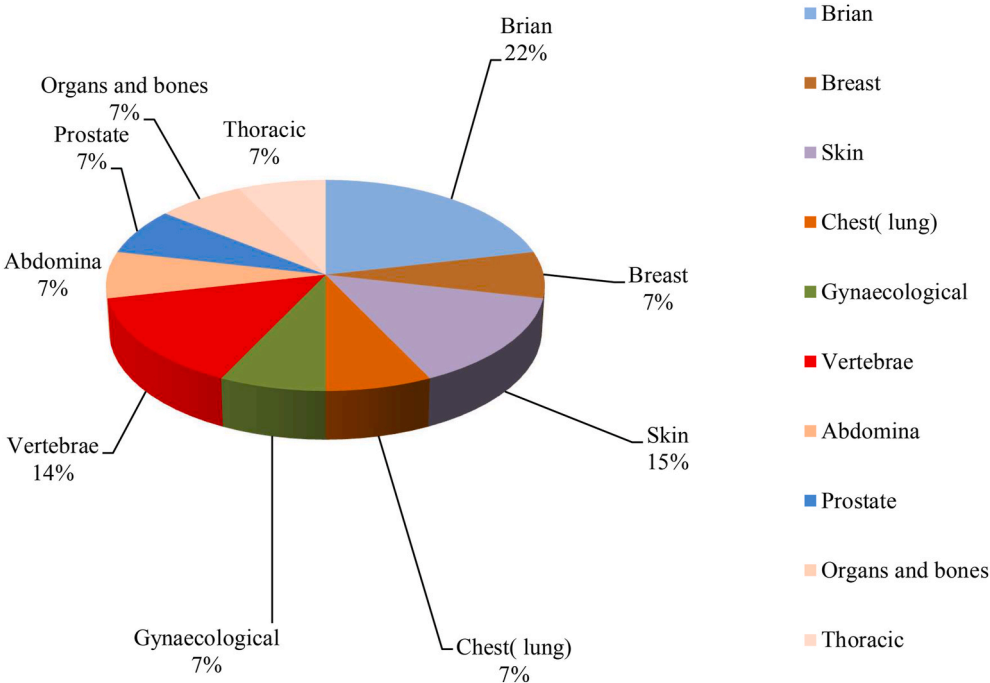


Fig. 20. Classification and proportion of medical image identification used in other image processing technology.